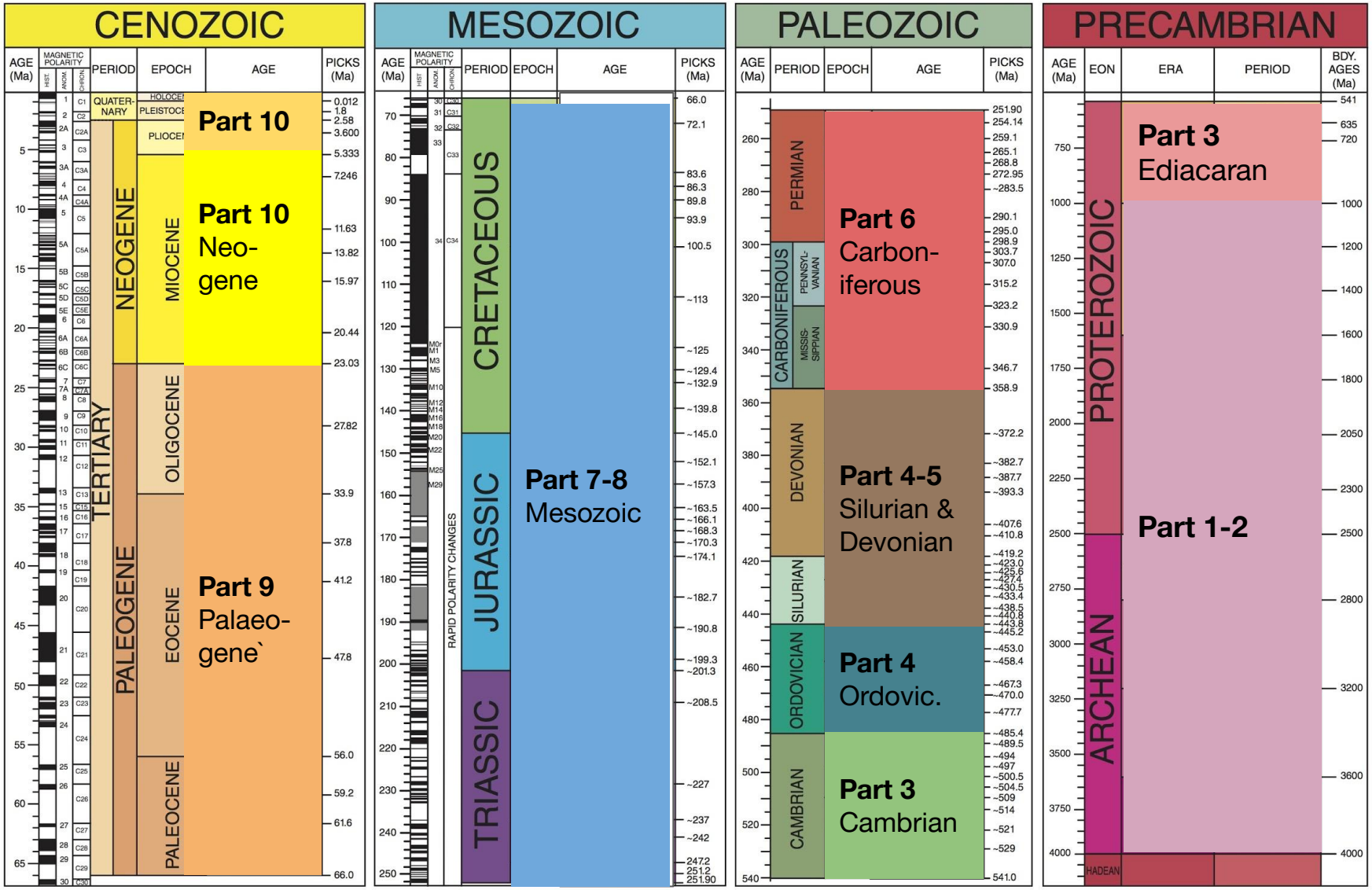
Part 9
Palaeogene, Dawn of the Modern

Birds, Mammals, and Fish

EART11100: How to Grow a Planet
Robert Sansom
University of Manchester



GSA GEOLOGIC TIME SCALE v. 5.0



Walker, J.D., Geissman, J.W., Bowring, S.A., and Babcock, L.E., compilers, 2018, Geologic Time Scale v. 5.0: Geological Society of America, https://doi.org/10.1130/2018.CTS005R3C.

©2018 The Geological Society of America

*The Pleistocene is divided into four ages, but only two are shown here. What is shown as Calabrian is actually three ages—Calabrian from 1.80 to 0.781 Ma, Middle from 0.781 to 0.126 Ma, and Late from 0.126 to 0.0117 Ma.

The Cenozoic, Mesozoic, and Paleozoic are the Eras of the Phanerozoic Eon. Names of units and age boundaries usually follow the Gradstein et al. (2012), Cohen et al. (2012) , and Cohen et al. (2013, updated) compilations. Numerical age estimates and picks of boundaries usually follow the Cohen et al. (2013, updated) compilation. The numbered epochs and ages of the Cambrian are provisional. A "~" before a numerical age estimate typically indicates an associated error of ±0.4 to over 1.6 Ma.

REFERENCES CITED

Cohen, K.M., Finney, S., and Gibbard, P.L., 2012, International Chronostratigraphic Chart: International Commission on Stratigraphy, www.stratigraphy.org (accessed May 2012). (Chart reproduced for the 34th International Geological Congress, Brisbane, Australia, 5–10 August 2012.)

Cohen, K.M., Finney, S.C., Gibbard, P.L., and Fan, J.-X., 2013, The ICS International Chronostratigraphic Chart: Episodes v. 36, no. 3, p. 199–204 (updated 2017, v. 2, http://www.stratigraphy.org/index.php/ics-chart-timescale; accessed May 2018).

Gradstein, F.M, Ogg, J.G., Schmitz, M.D., et al., 2012, The Geologic Time Scale 2012: Boston, USA, Elsevier, https://doi.org/10.1016/B978-0-444-59425-9.00004-4.

Previous versions of the time scale and previously published papers about the time scale and its evolution are posted to http://www.geosociety.org/timescale.

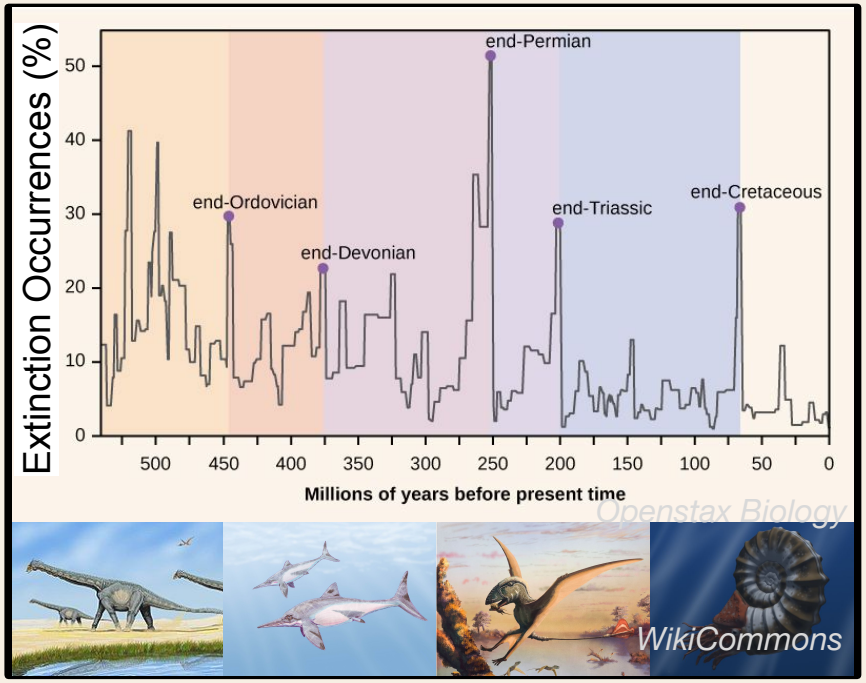
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Mesozoic Summary

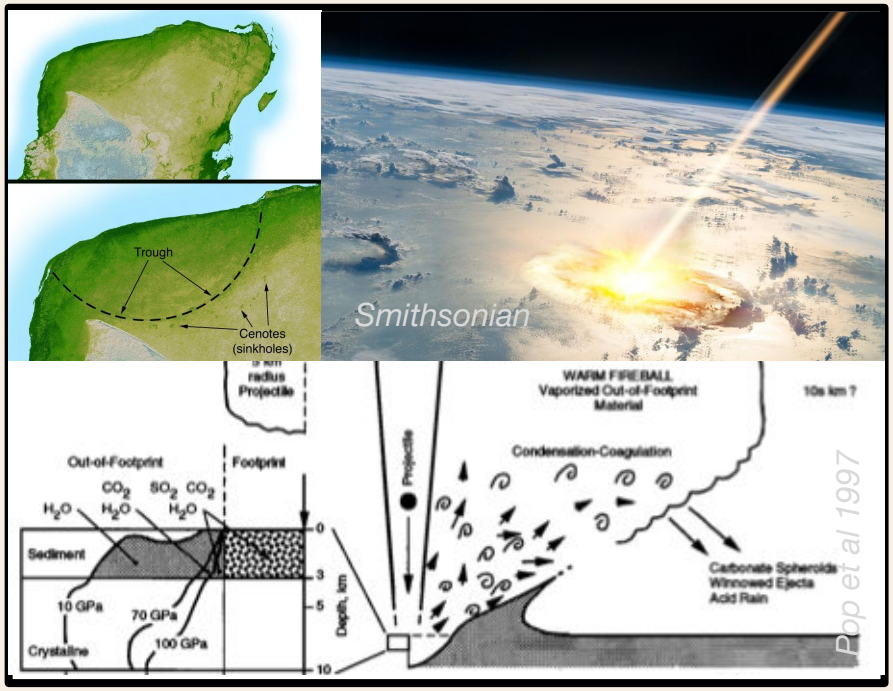
- Major groups of archosaurs originate in the aftermath of the Permian-Triassic mass extinction event (the big one)
- Vertebrates diversify and generally grow in body size in the Jurassic and Cretaceous whilst the climate becomes more humid and plants also diversify and spread
- The Mesozoic ends with an impact-related extinction



Dinosaur Extinction

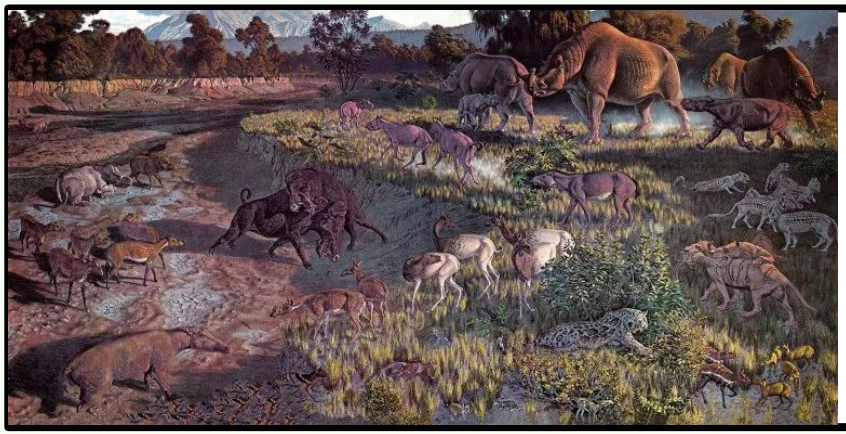


Enormous amount of faunal change at the end of the Cretaceous: end of dinosaurs, ichthyosaurs, pterosaurs, and ammonoids



Massive extra terrestrial impact was likely cause of the massive **Cretaceous-Palaeogene extinction**

The Post Extinction Palaeogene



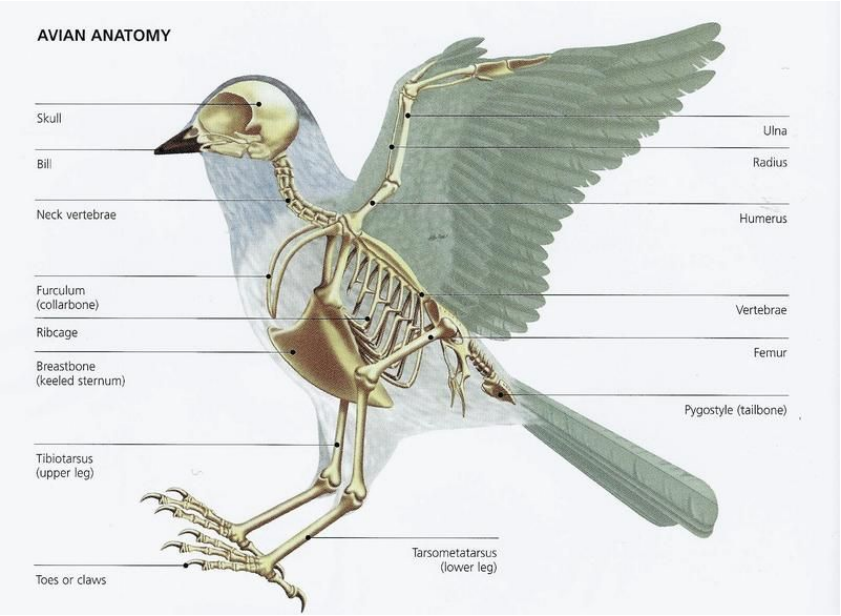
Very different flora, fauna, ecosystems, and climate after the extinction



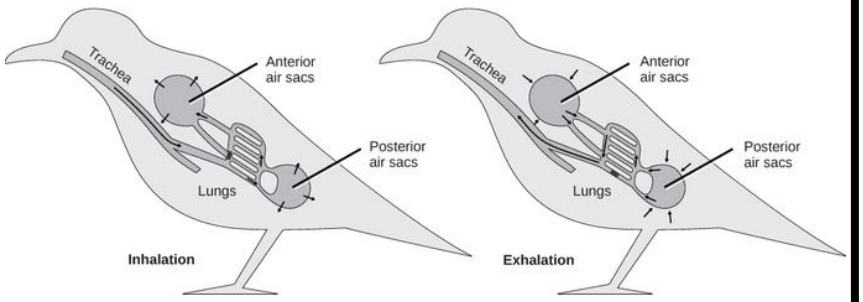




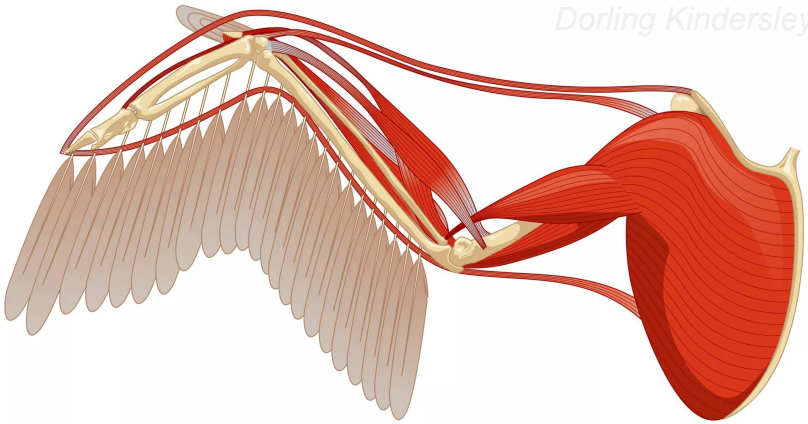
Birds



Birds (**Aves**) are specialized archosaurs, defined by: Feathers, Endothermy, Hollow bones, Toothless Beak, Clasping claws

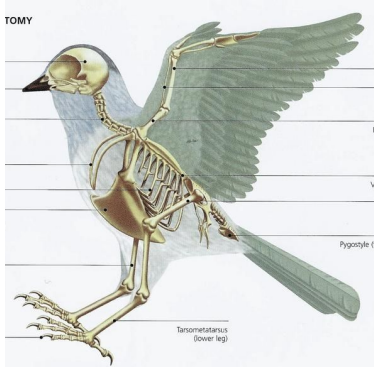


Complex air breathing through series of air sacs with bidirectional flow

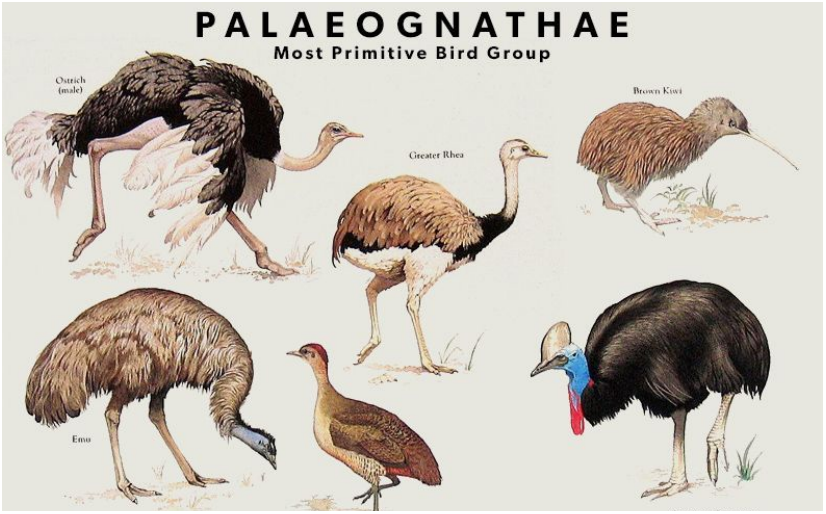


Pectoral appendage modified for flight: large muscles, reduced manus

Bird Diversity



Birds are highly diverse - about 10,000 species



Flightless Palaeognathae, 47 species



Fowl (Galloanserae), ~400 species



Neoaves(most birds), +9000 species

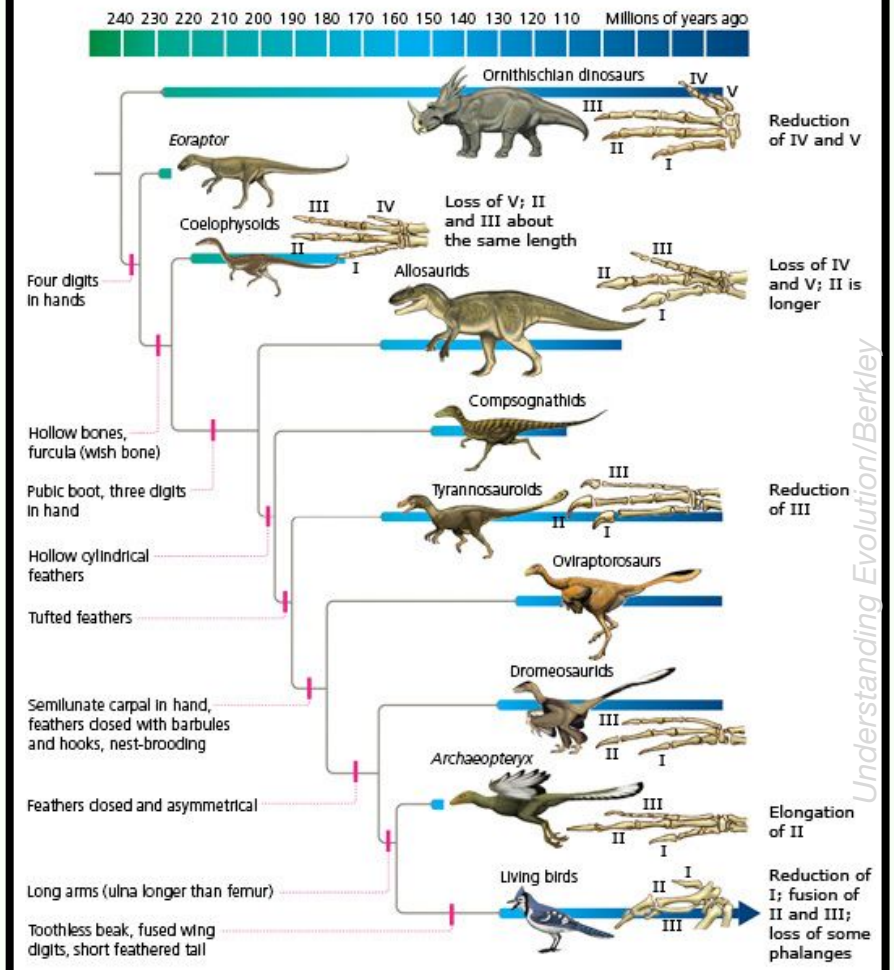
Birds are Theropods





The birds evolved from within the **theropod** dinosaur group:

Dinosaurs aren't extinct?



Step-wise acquisition of avian characters along the avian stem

Evolution of Feathers

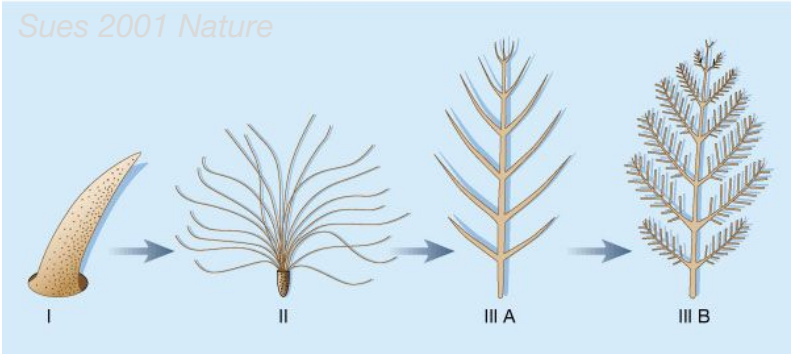


Archaeopteryx (Jurassic)

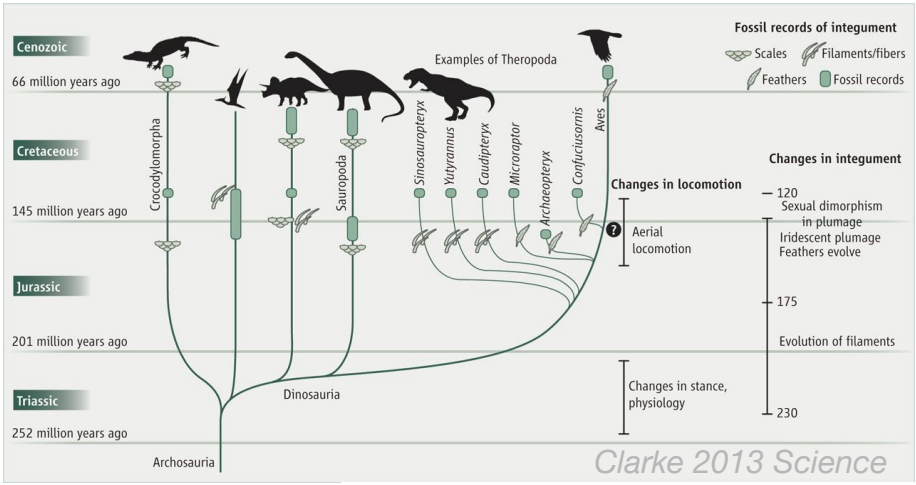


New Scientist

Archaeopteryx (Jurassic stem-bird) has well-developed flight feathers



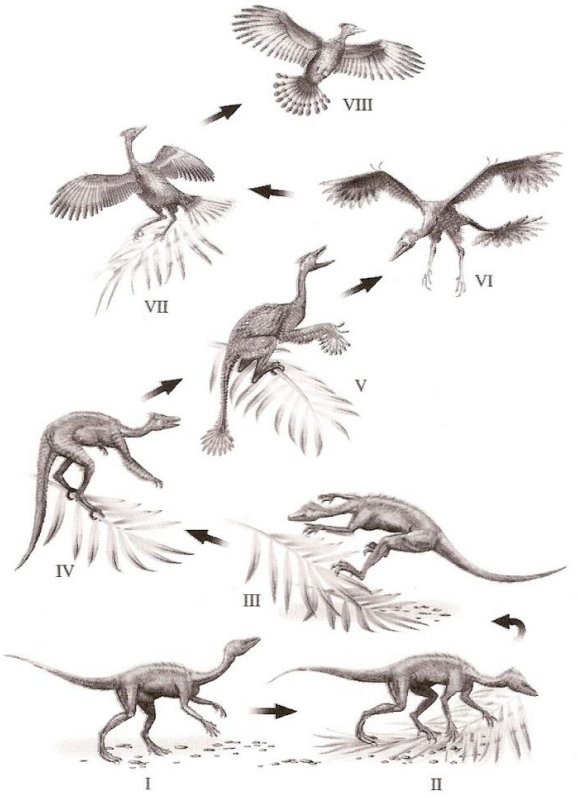
Hollow filaments, tufts, serial barbs, secondary barbs



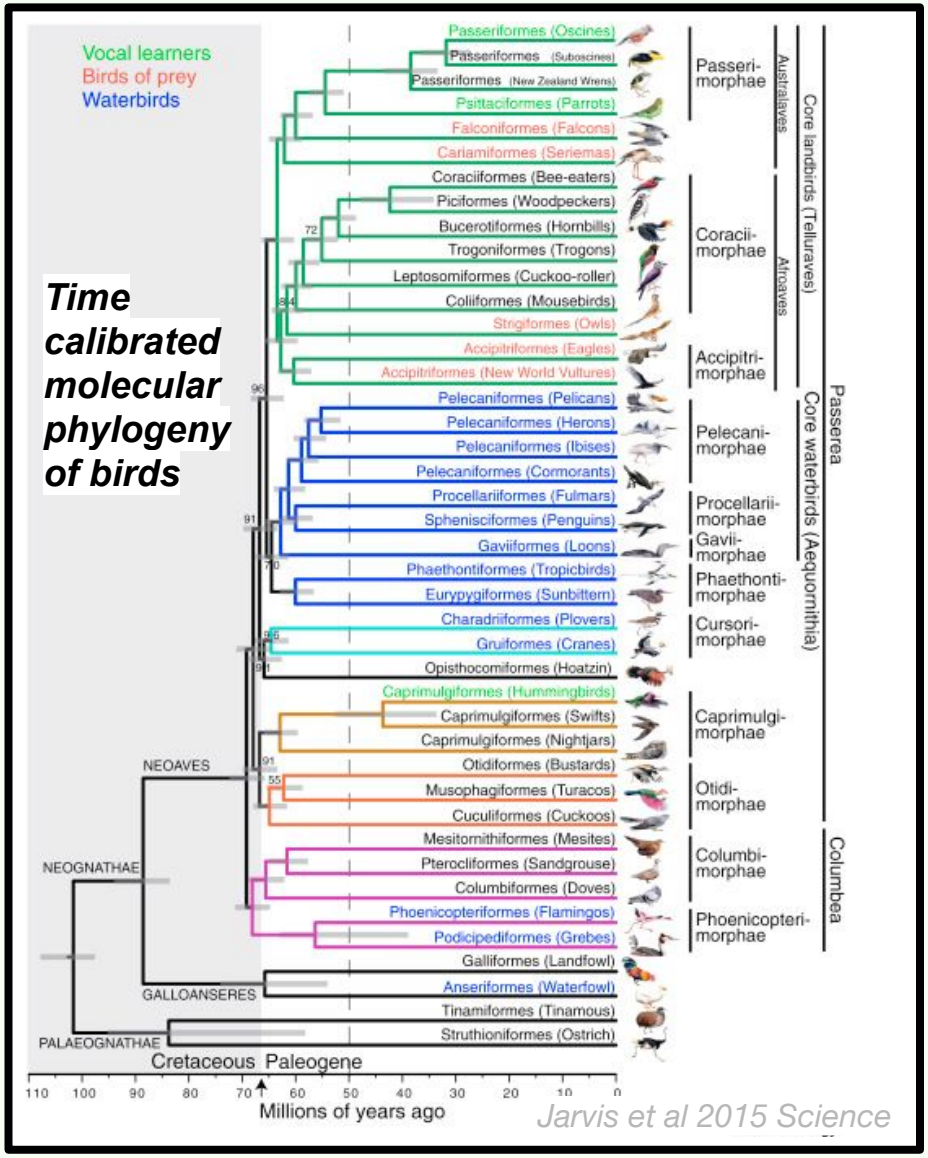
Feathers have a long evolutionary history

Filamentous barbs seen in pterosaurs, dinosaurs

Feathers with barbules seen in *Microraptor*, Avialae

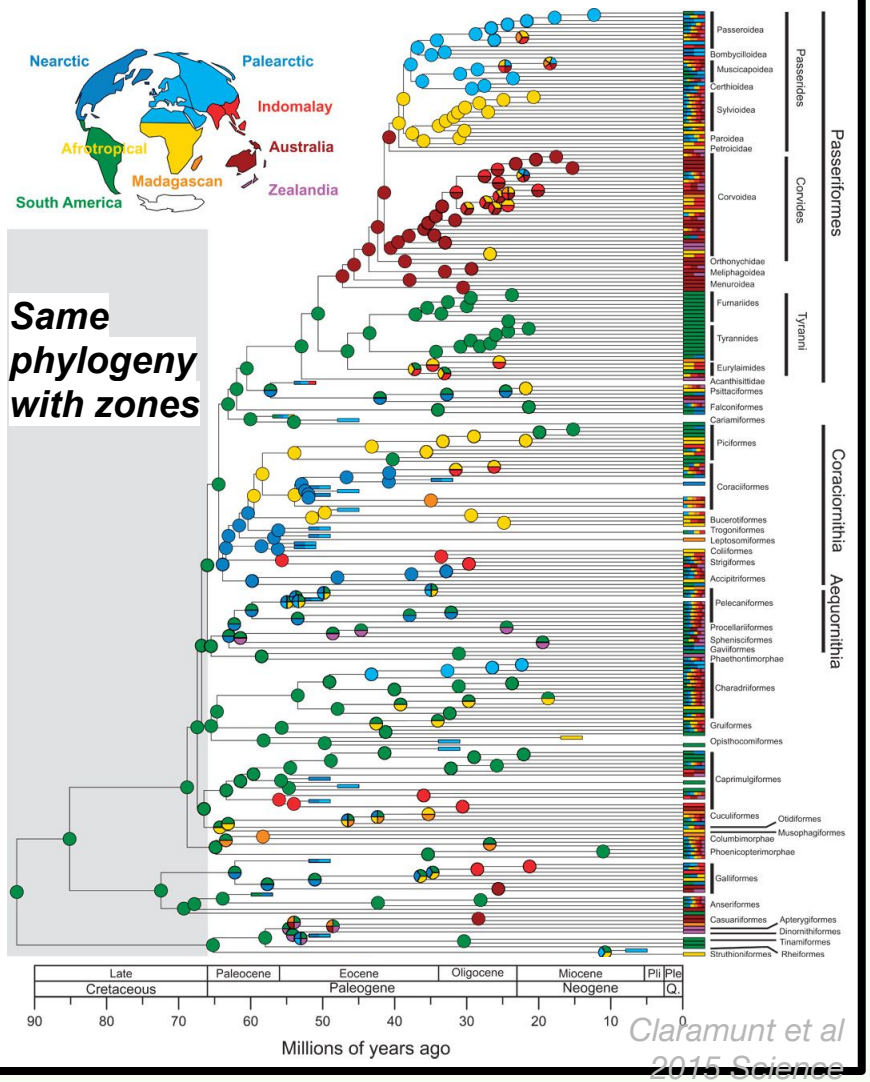


Timing of Bird Evolution



Jarvis et al 2015 Science

Birds (esp. Neoaves) underwent massive radiation just after the **Cretaceous-Palaeogene Extinction**
Diversify in ecology and across globe



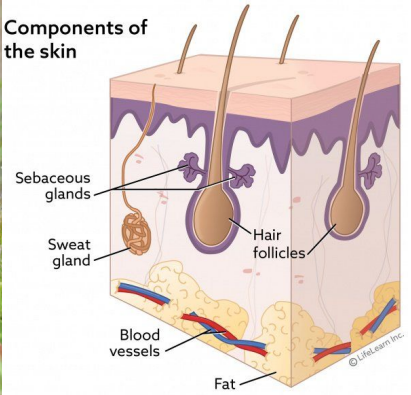
Claramunt et al 2015 Science

Mammals

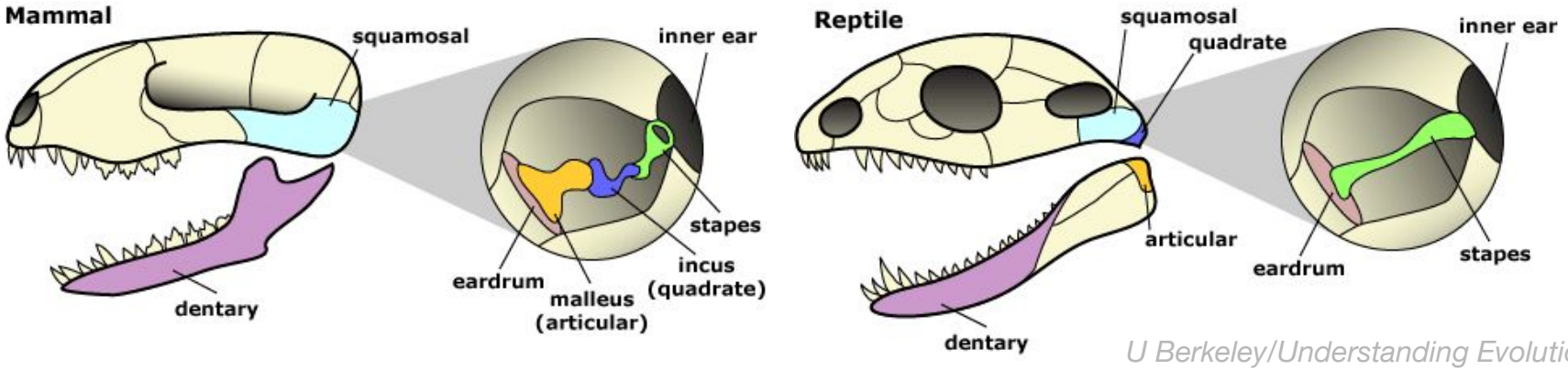


Hair and Sweat Glands





Just one jaw bone (**dentary**), three middle ear bones



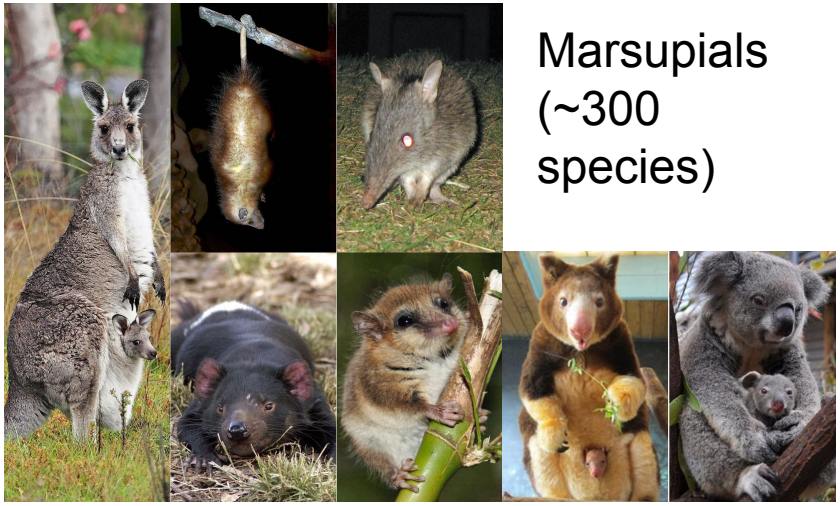
U Berkeley/Understanding Evolution

Mammals Diversity

Monotremes (5 species)



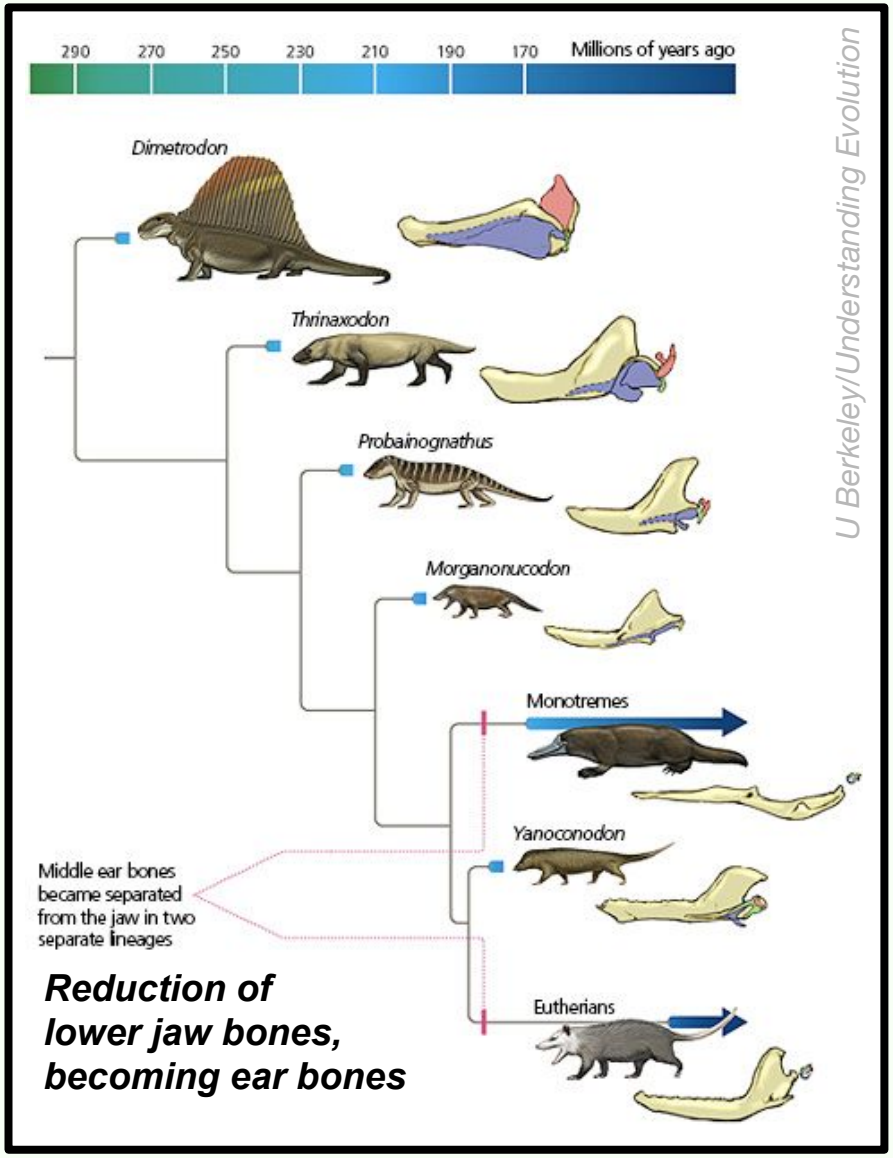
Marsupials (~300 species)

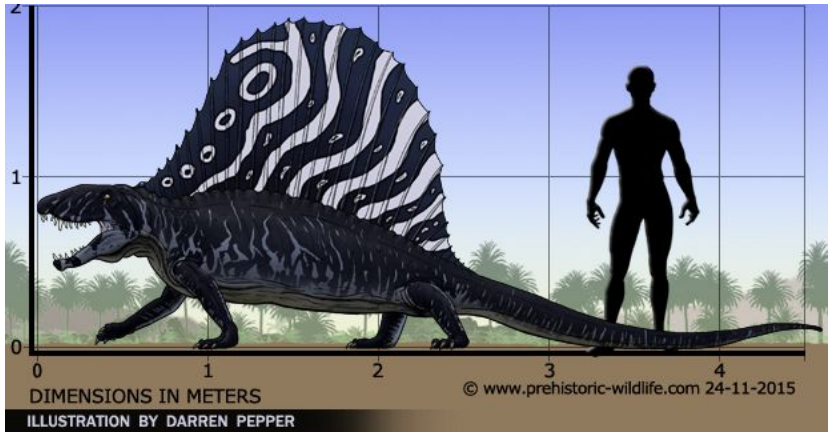


Placentals (~6000 species)



Mammal Origins



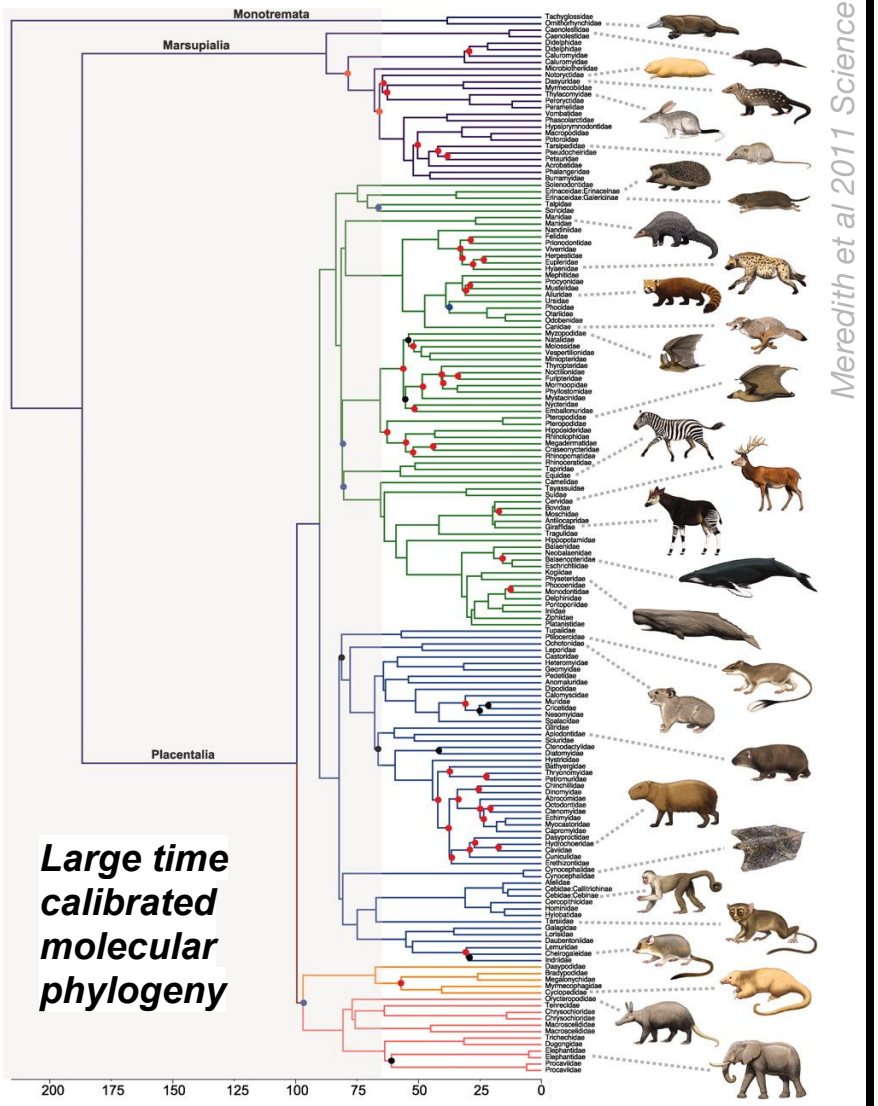


Early Synaspids included *Dimetrodon* with large sail from Early Permian

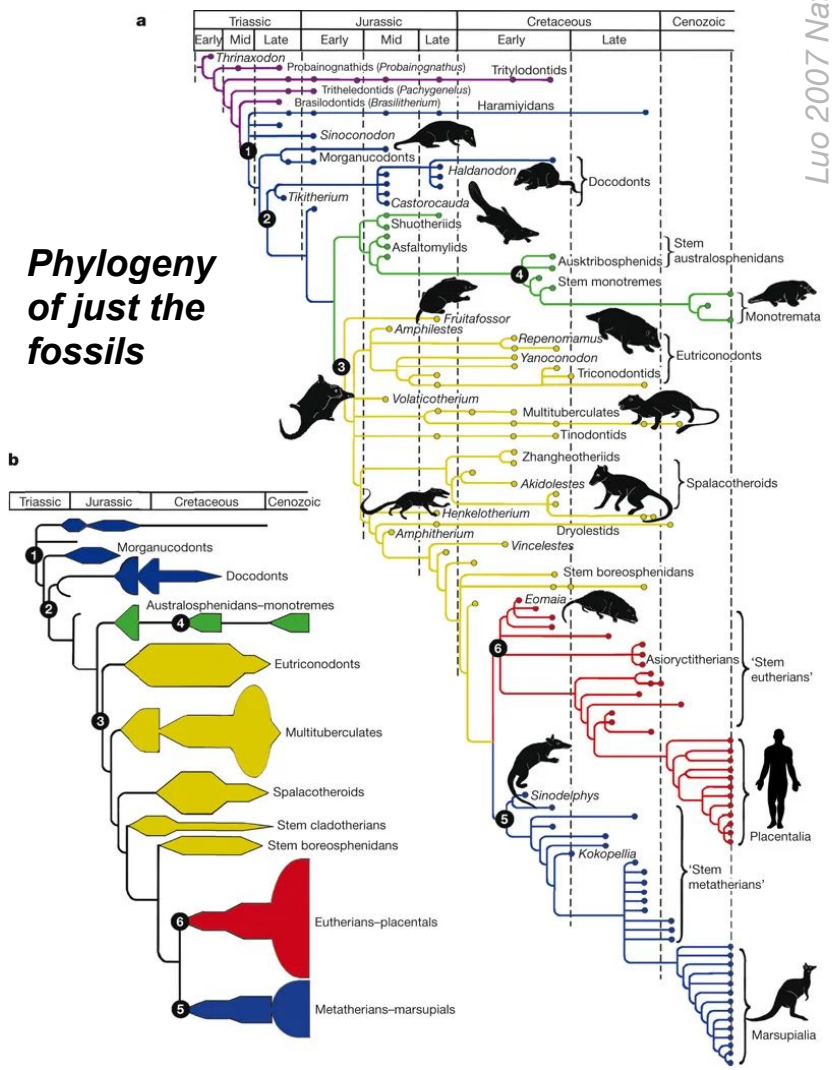


The stem-mammal *Morganucodon* from Late Triassic had deciduous tooth shedding, probably hair

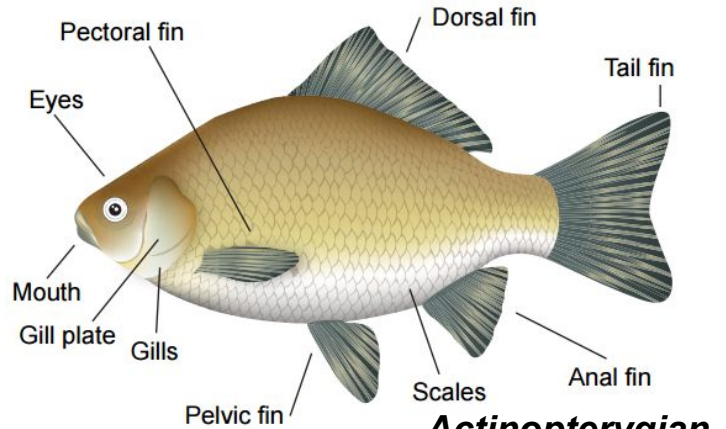
Mammal Evolution



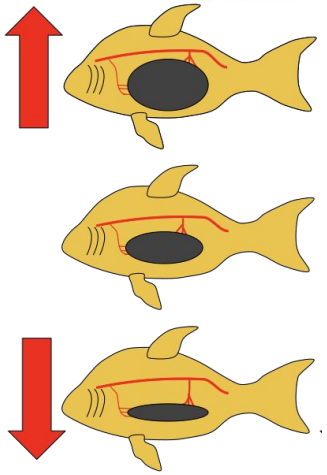
Most placental mammals radiated after the Cretaceous Palaeogene Extinction Event, some lineages originating slightly before



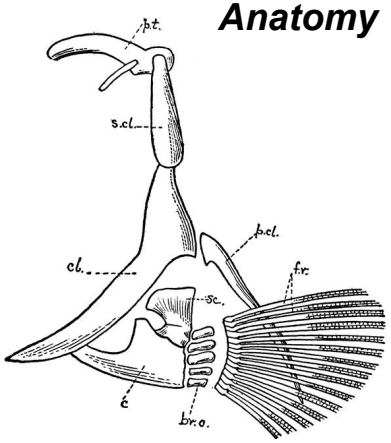
Ray-finned fishes (Actinopterygii)



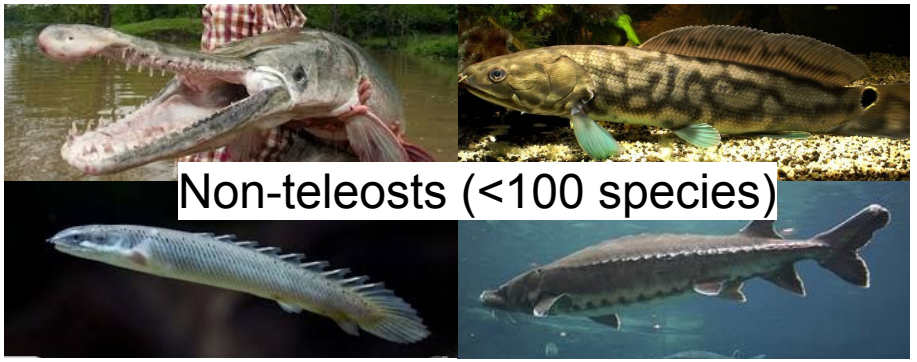
Actinopterygian Anatomy



Air filled swim bladder for buoyancy



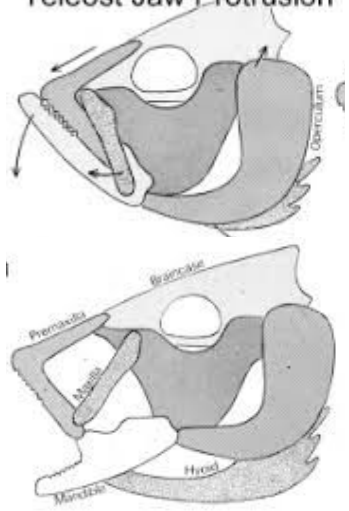
Fin rays (radials) attached directly to base



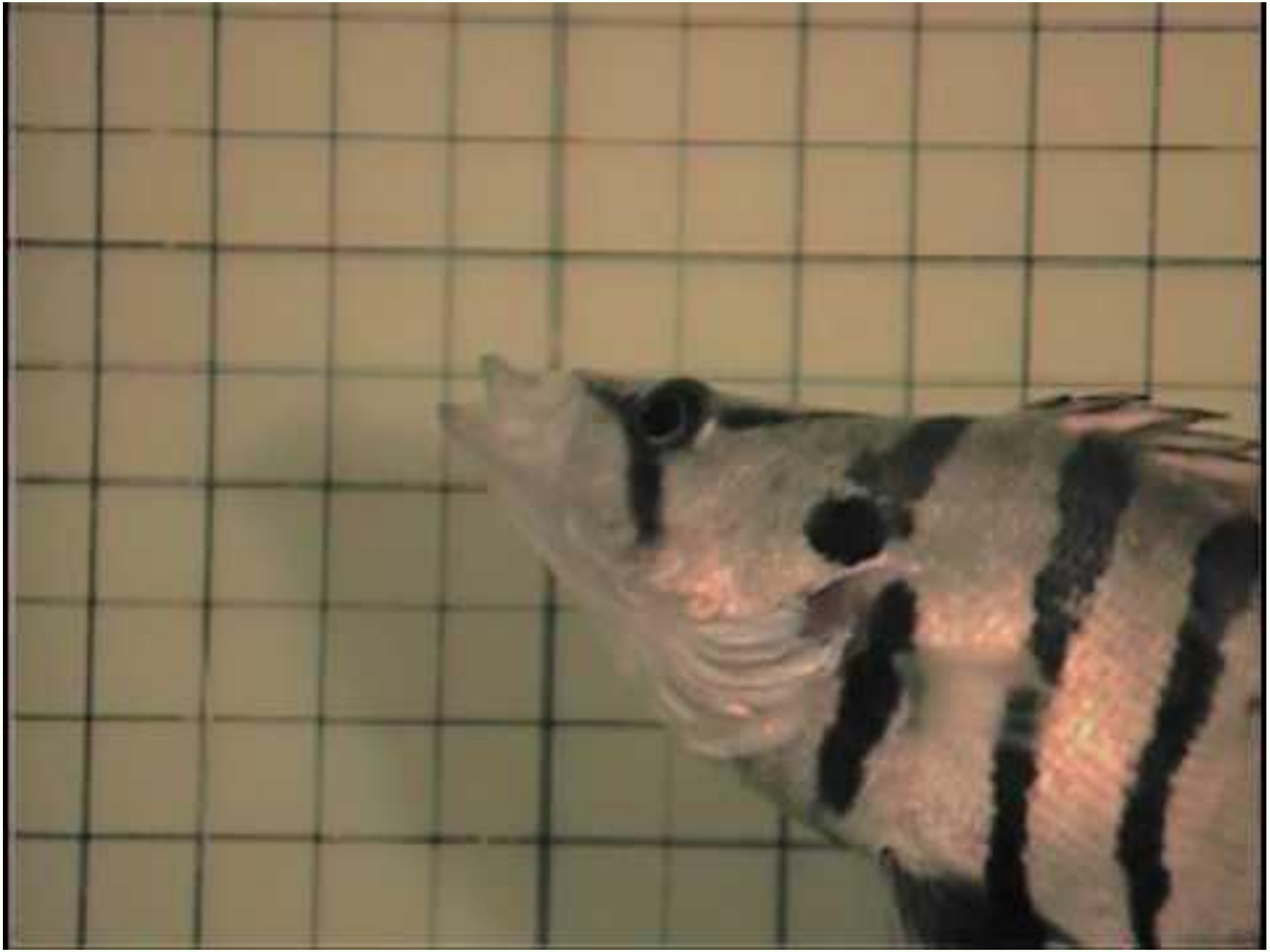
Non-teleosts (<100 species)

Teleosts <26,000 species

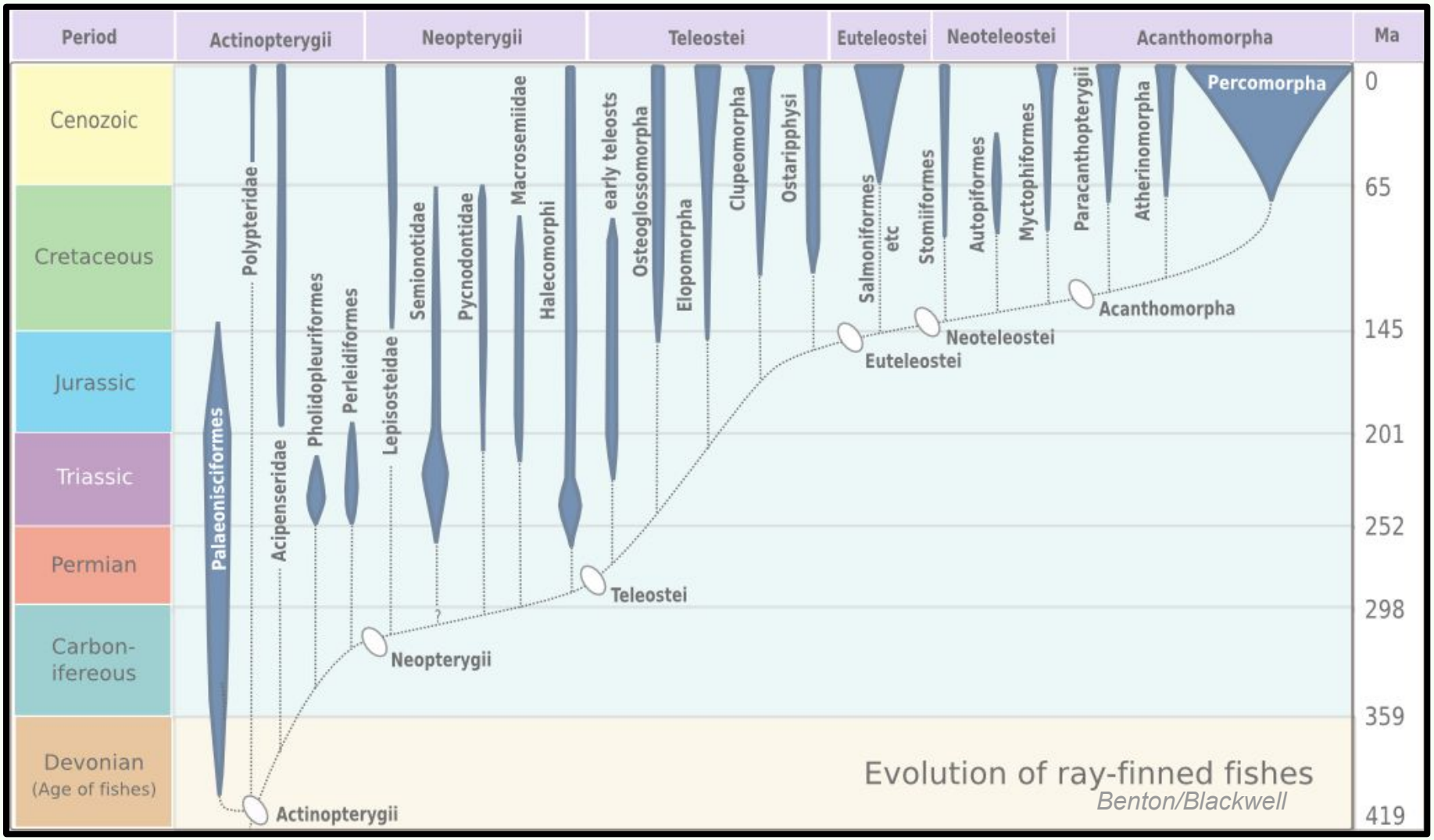
Teleost Jaw Protrusion







Teleost Evolution



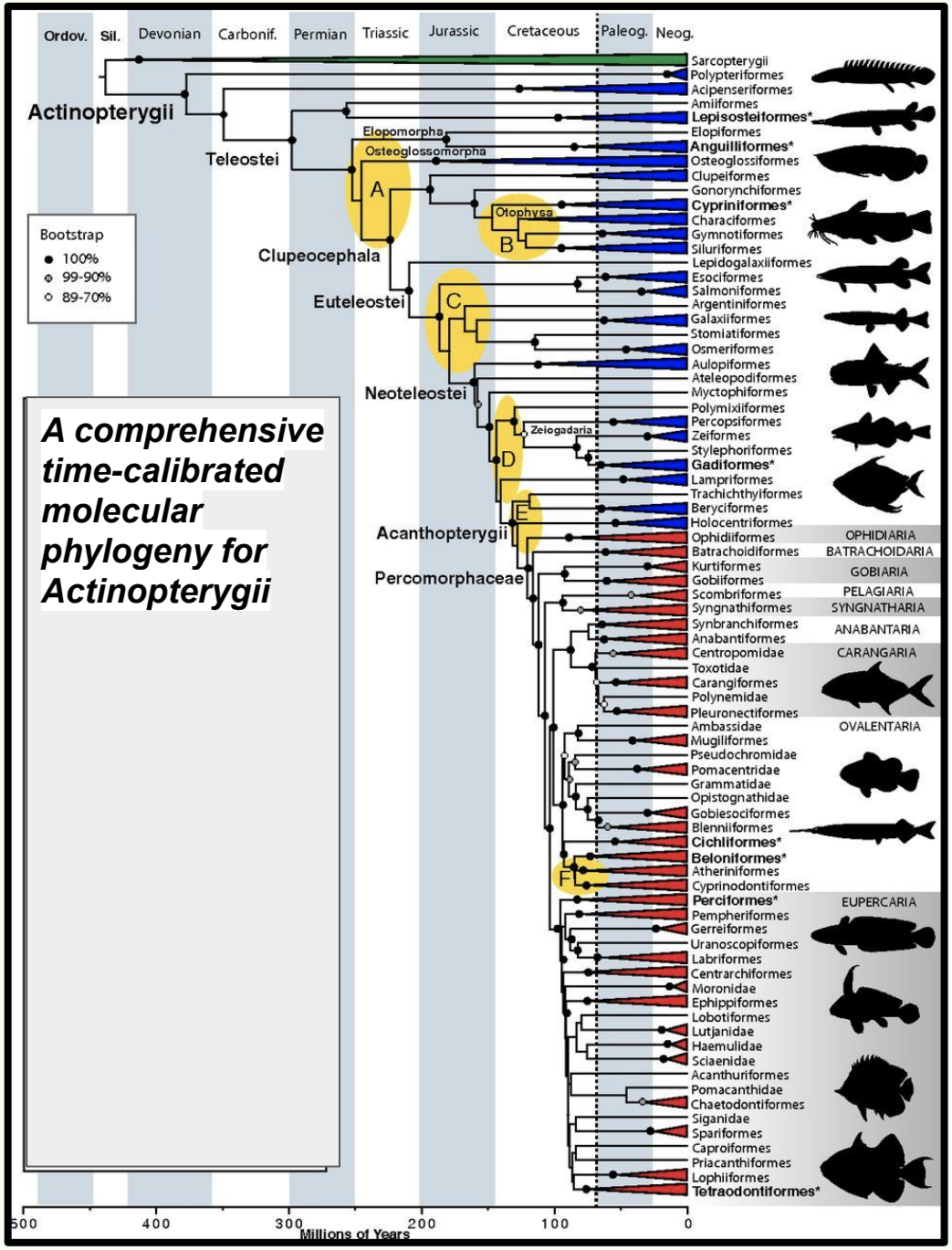
Teleost Evolution

Teleosts diversified in the Mesozoic, before the end Cretaceous extinction

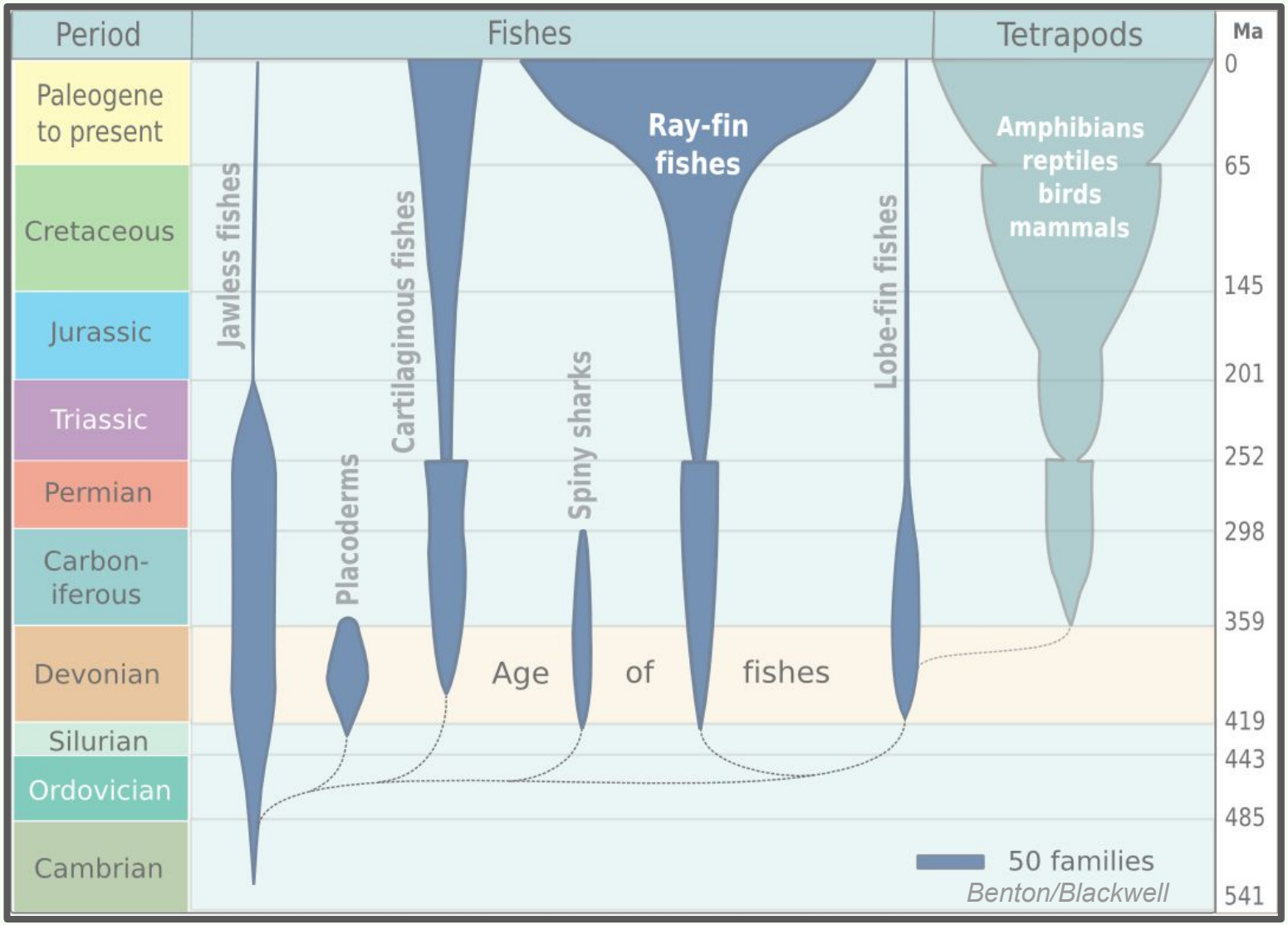
Many sub-clades underwent rapid diversification after the Cretaceous in the Palaeogene

Less pronounced pattern than for birds and mammals

Hughes et al 2018 PNAS



Palaeogene Vertebrate Diversifications



Palaeogene, Dawn of the Modern Birds, Mammals, and Fish

Intended Learning Objective: Define and describe birds, mammals, actinopterygians in terms of their anatomy and diversity

Take a look at the associated task/question

Robert Sansom, U. Manchester



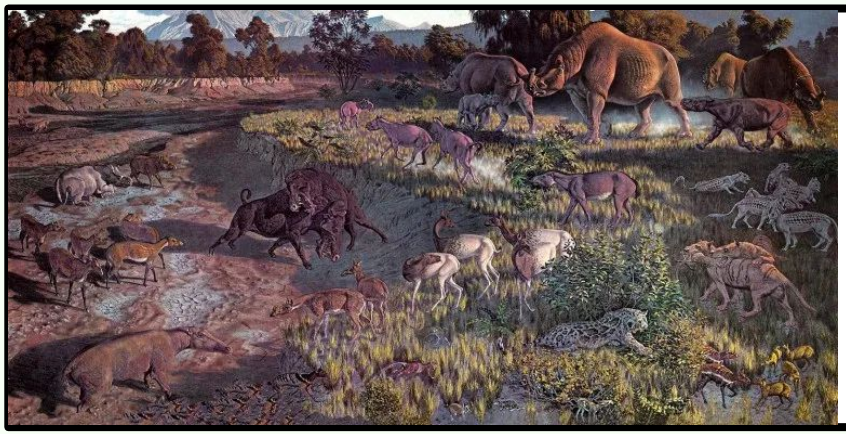
Part 9
Palaeogene, Dawn of the Modern

Adaptive Radiation

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The Post Extinction Palaeogene



Very different flora, fauna, ecosystems, and climate after the extinction



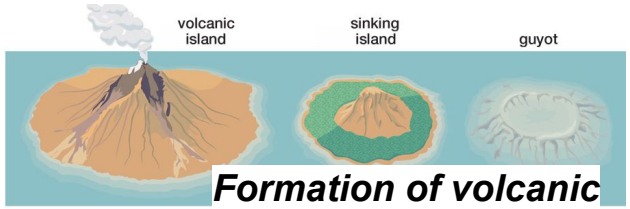


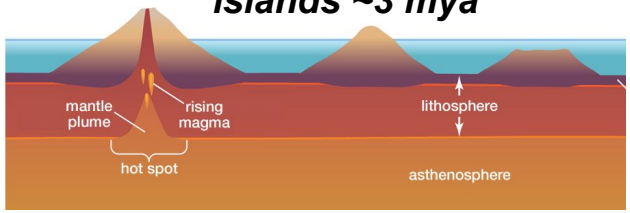


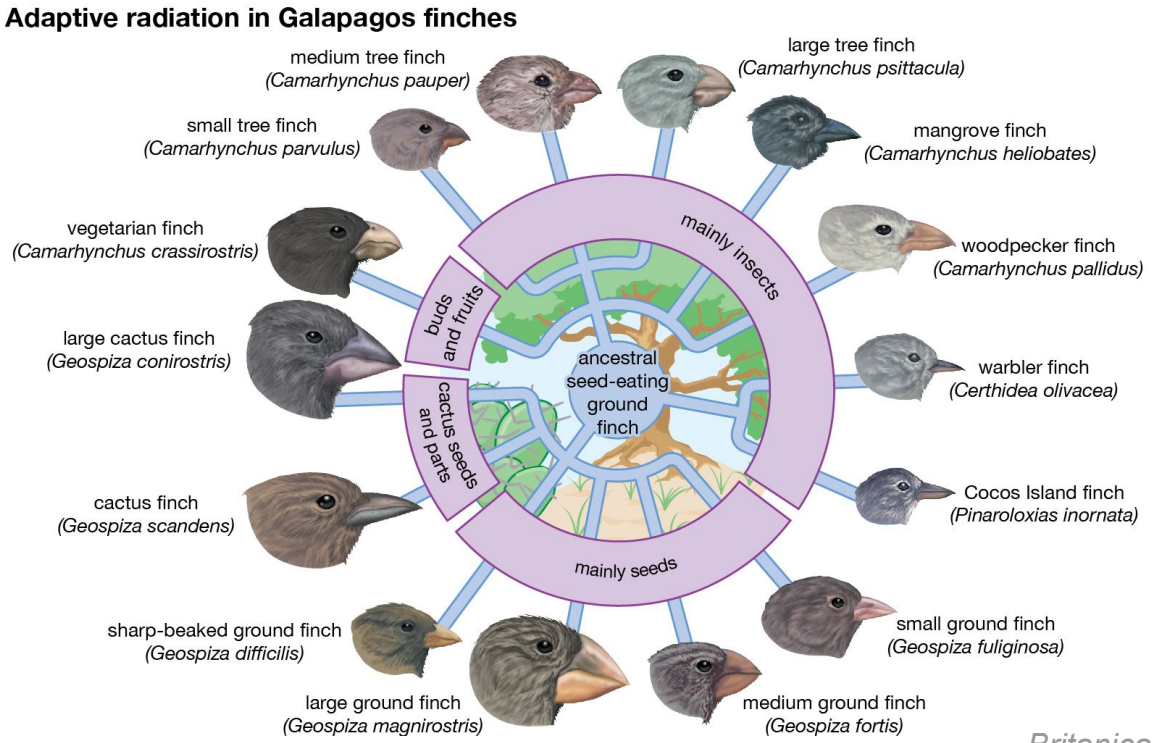
Adaptive Radiation

An **adaptive radiation** is a rapid diversification of organisms from an ancestral species into a multitude of new forms, particularly to occupy newly available **niches**
A good example is the adaptive radiation of Darwin's **Galapagos Finches**



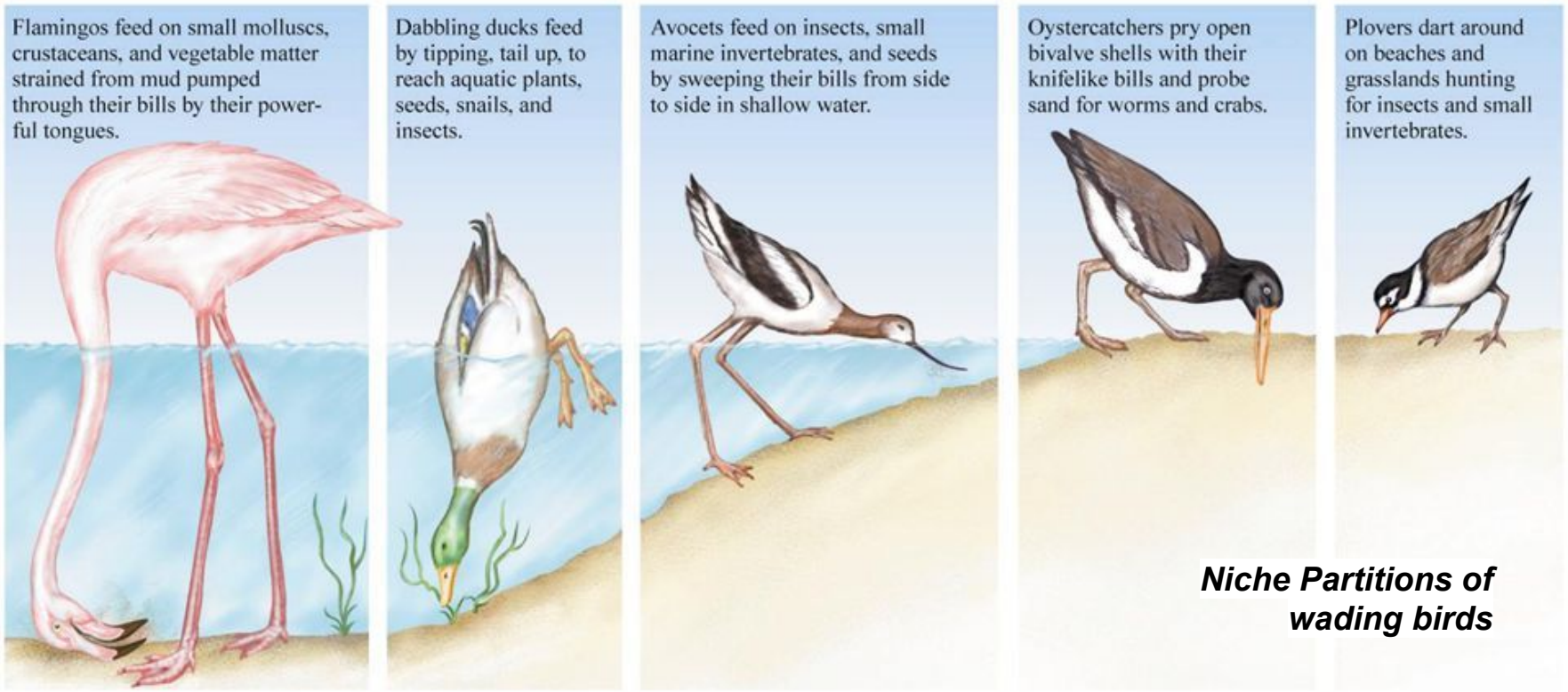






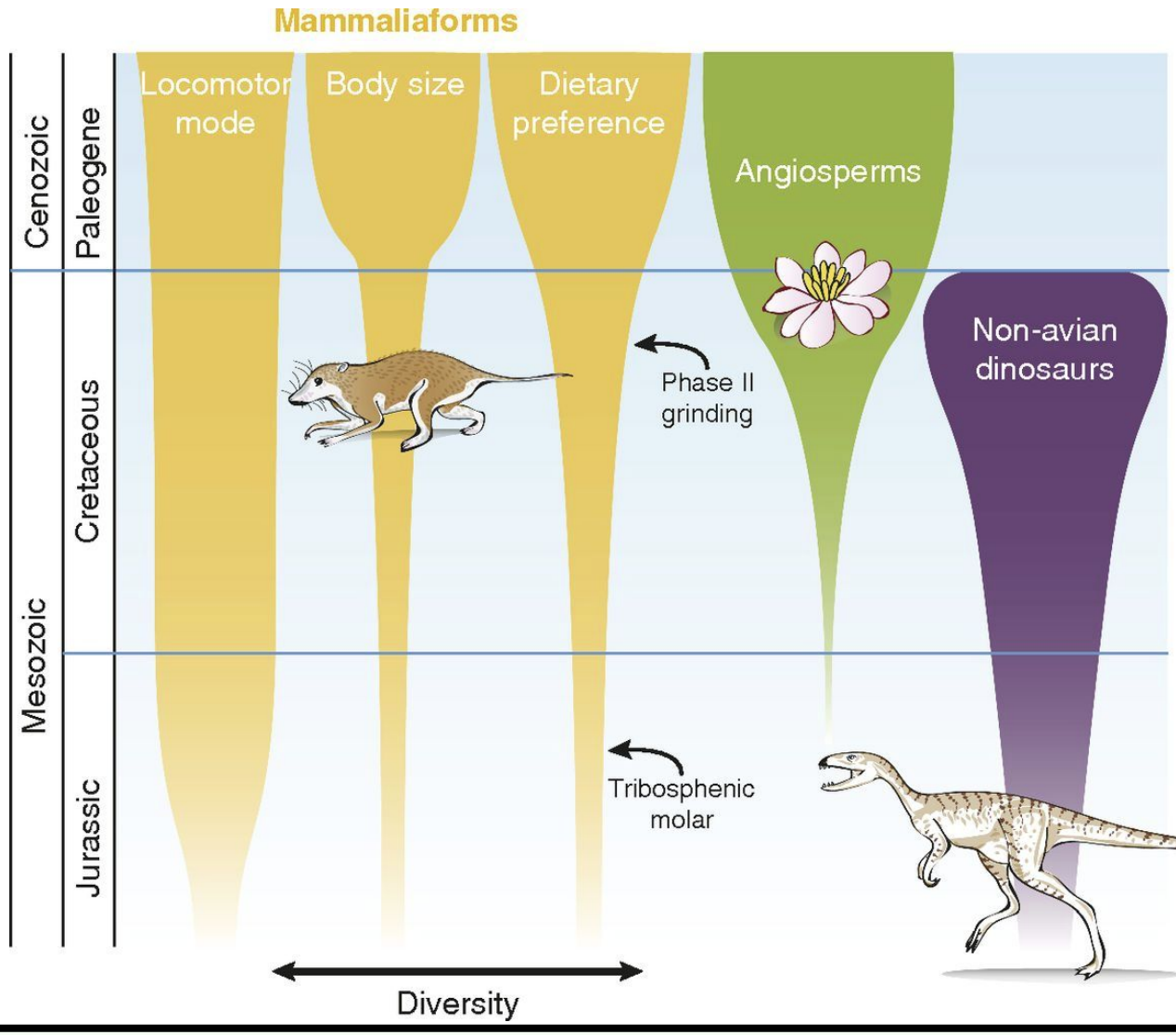
Ecological Niche

Each organism has an **ecological niche**: the *position* of a species within an ecosystem in terms of the range of conditions it can persist in
This includes the **biotic environment** (interactions with other species such as competitors, predators, etc) and **abiotic environment** (e.g. temperature, O_2 etc).



Niche Partitions of wading birds

Mass extinction causes niche availability

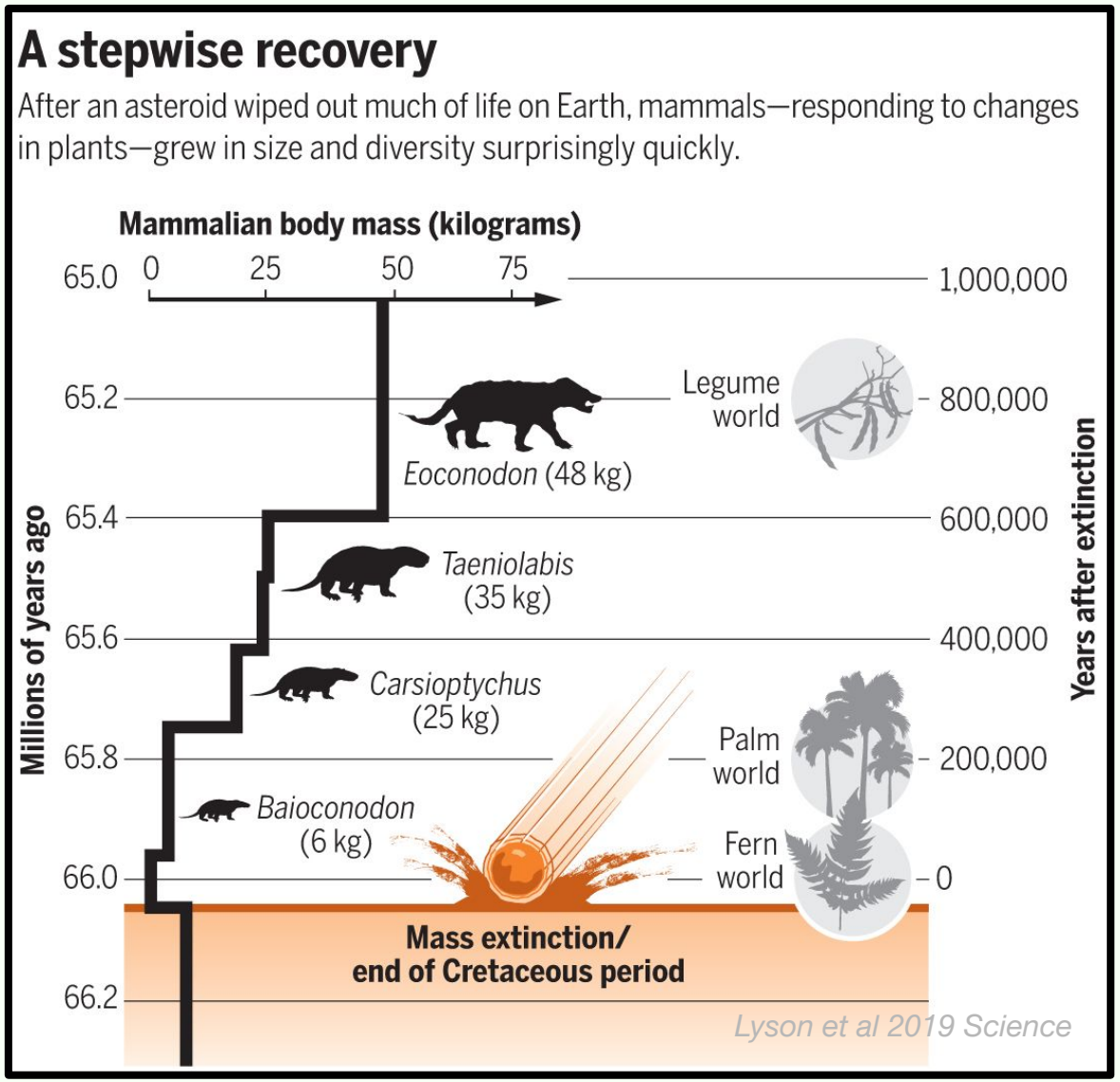


After the extinction of the dinosaurs, mammals were able to radiate to occupy newly available niches, particularly large body sizes

Also mammalian specific innovations (teeth and grinding) and new biological niches from radiation of angiosperms

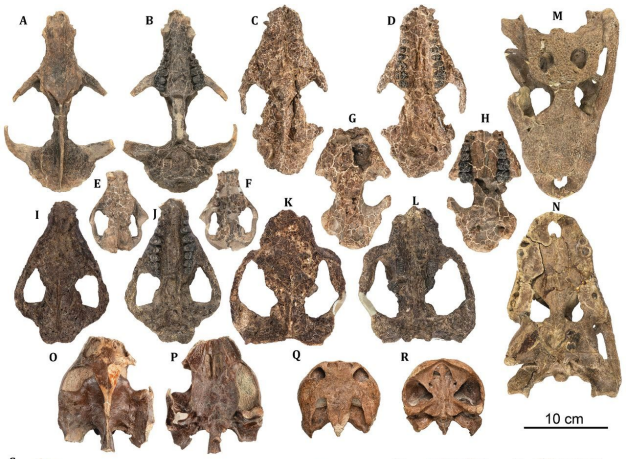
Chen et al 2019 PNAS

Rapid Diversification of Mammals

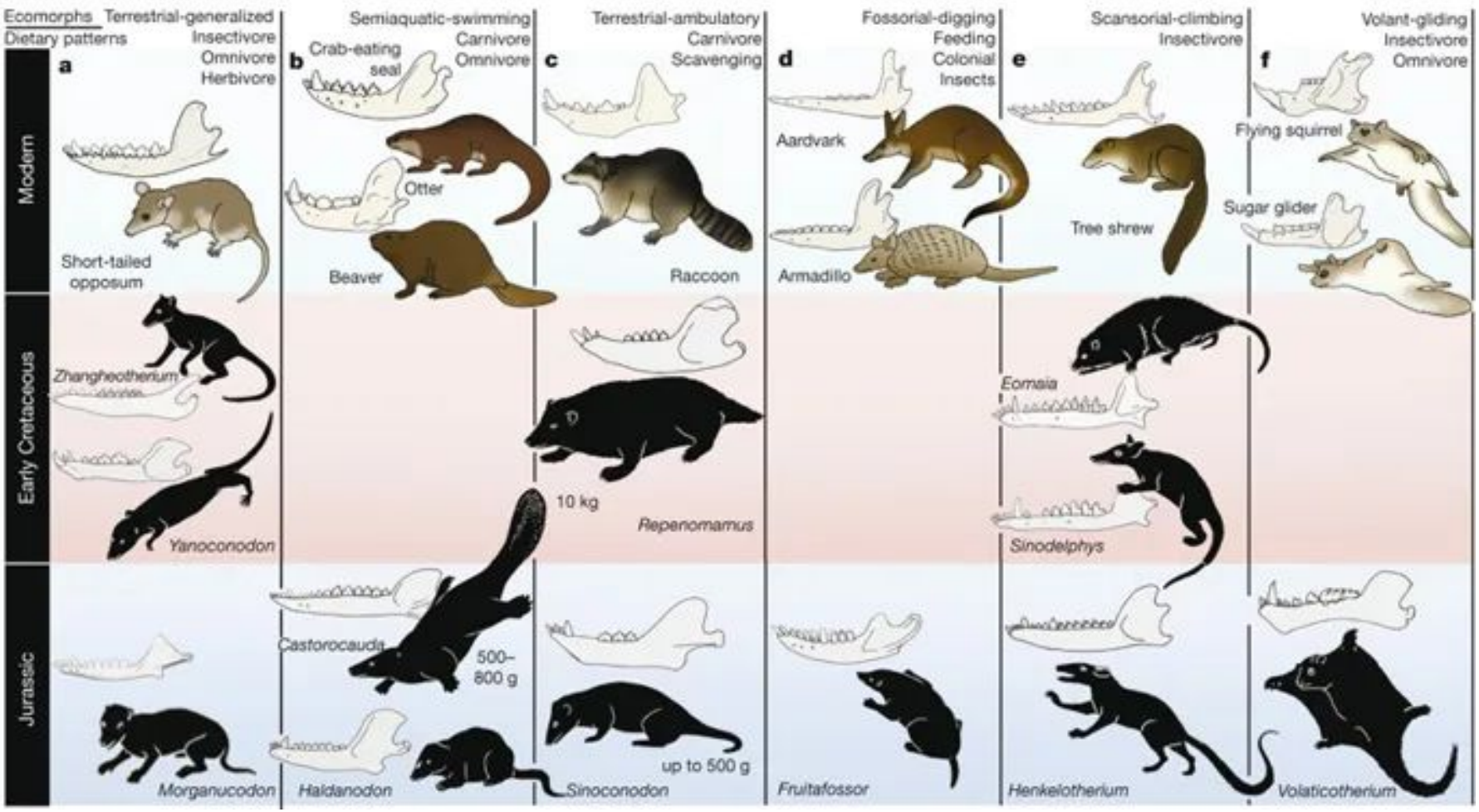


New site in Colorado with complete sedimentary record before and after extinction, vertebrate and plant fossils, volcanic ash layers to date

Plants recovered very quickly, mammal body size got a lot bigger very quickly



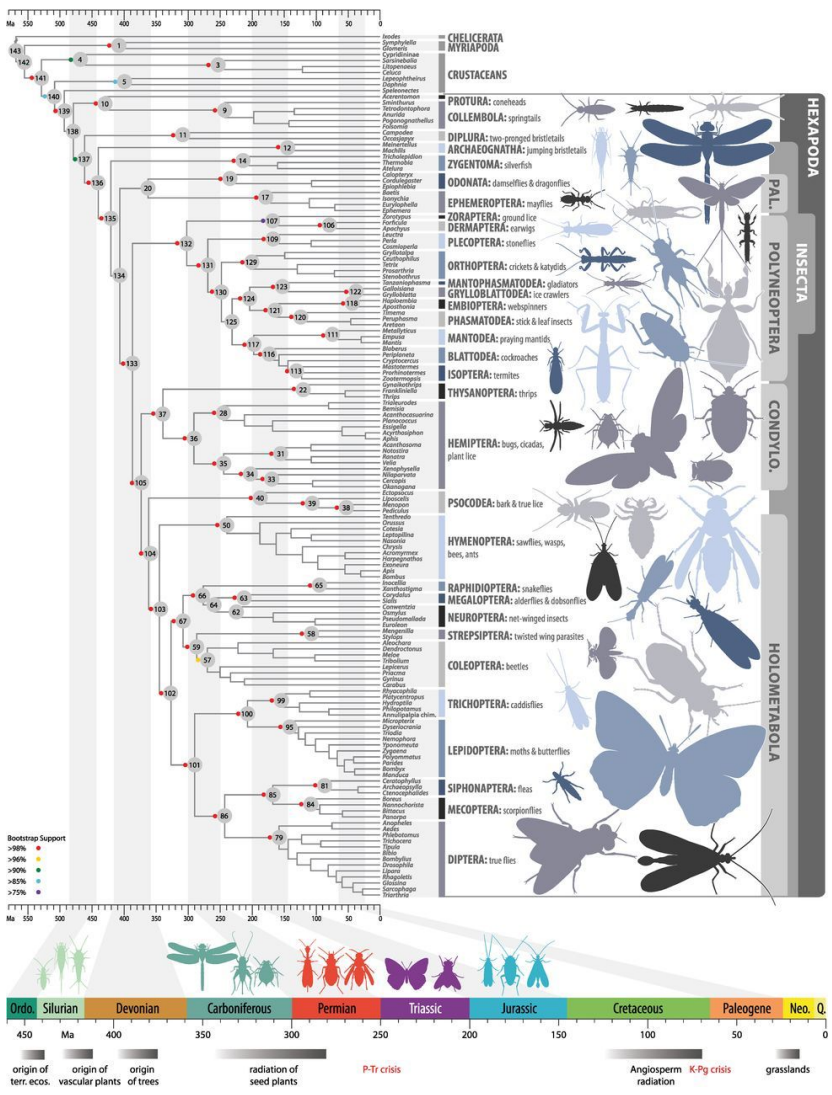
Niche Replacement in Mammals

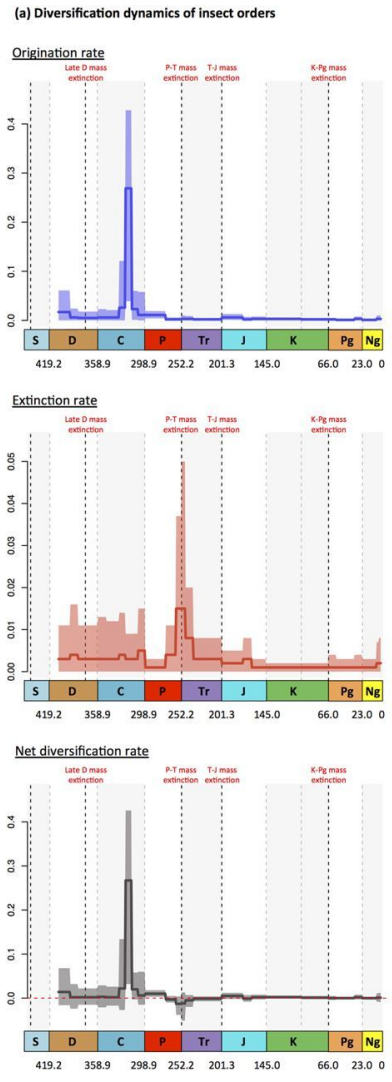


Modern and Mesozoic niches occupied by different small mammals

Luo 2007 Nature

Insects do their own thing





Insect diversification takes place in the Carboniferous

They suffer big losses at the Permian-Triassic extinction event

Relatievly unaffected by the Cretaceous-Palaeogene extinction event

Misof et al 2014 Science
Condamine et al 2016 Scientific Reports

Palaeogene, Dawn of the Modern. Adaptive Radiation

Intended Learning Objective: Explain adaptive radiations in terms of evolutionary dynamics and niche occupations

Now take a look at the associated task/discussion

Robert Sansom, U. Manchester



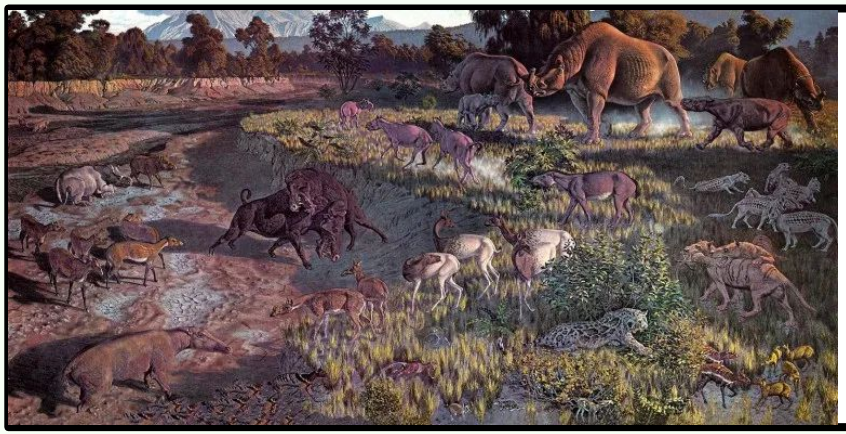
Part 9
Palaeogene, Dawn of the Modern

Climate and Evolution

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Robert Sansom
University of Manchester



The Post Extinction Palaeogene



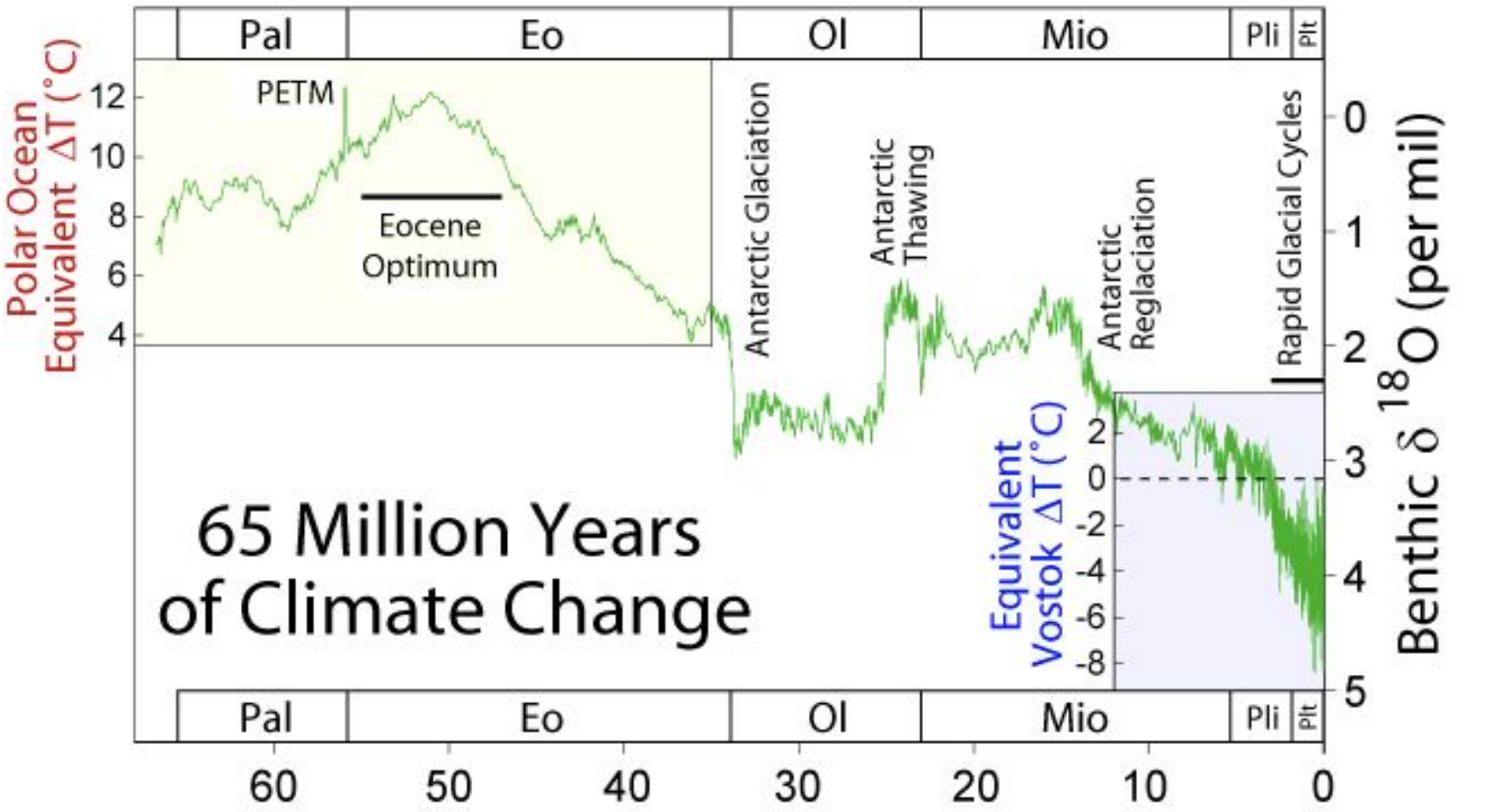
Very different flora, fauna, ecosystems, and climate after the extinction





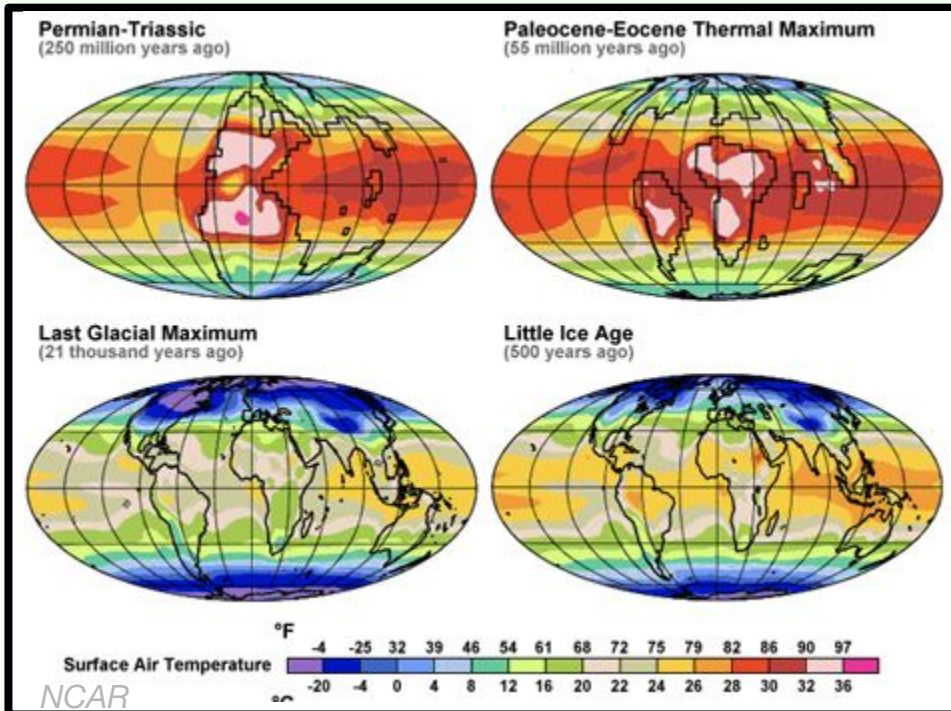


Climate Changes



The palaeogene starts with high global temperatures with little polar ice, marked by the **Palaeocene-Eocene thermal maximum (PETM)**

Palaeocene-Eocene Thermal Maximum



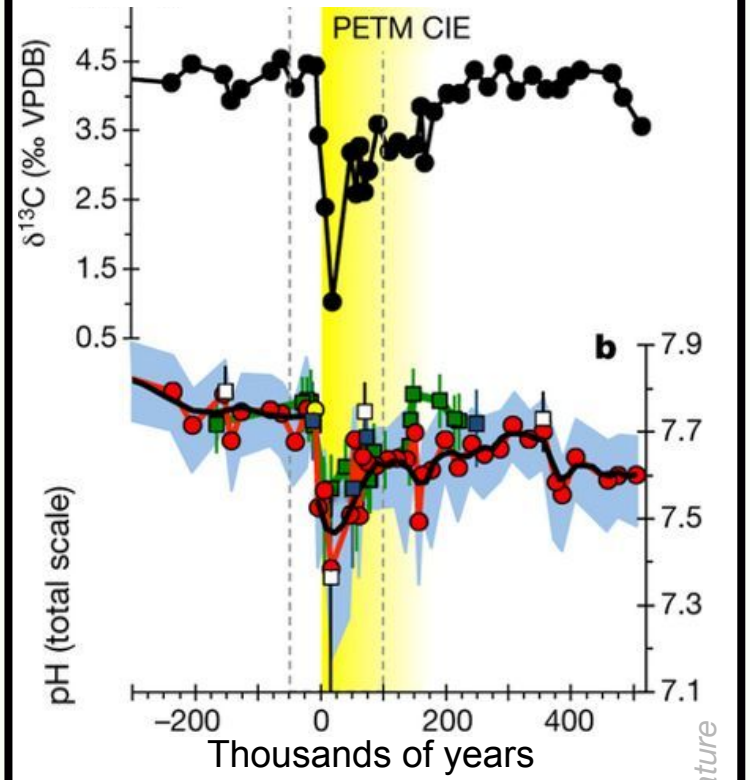
Palaeocene-Eocene thermal maximum (**PETM**) temperatures were about 5-8º higher (greenhouse world) than today (ice house world)

Massive injection of atmospheric CO_2 over a short period of time.
Possible causes include volcanism, comet impacts, and their effects on carbon cycles and ocean circulation



Increasing abundance and spread of mammals on land.
In deep oceans, extinction of benthic foraminifera (primary consuming plankton with calcareous ‘test’)

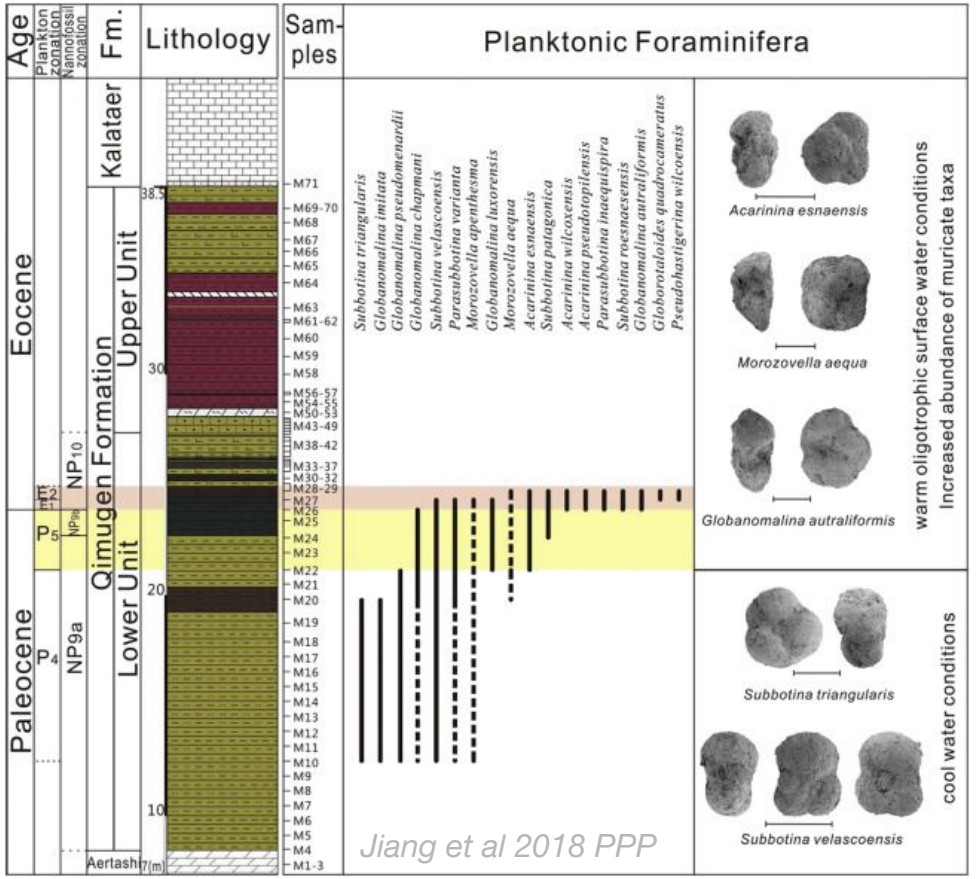
Palaeocene-Eocene Thermal Maximum



Volcanism caused spike in oceanic CO_2, drop in pH over 1000s years

Earth recovery aided by enhanced burial carbon

Gutjahr et al 2017 Nature



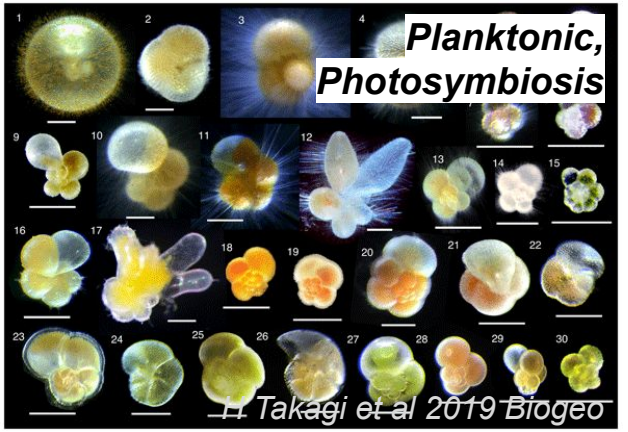
Extinction of benthic foraminfera related to acidification (calcareous test) at PETM

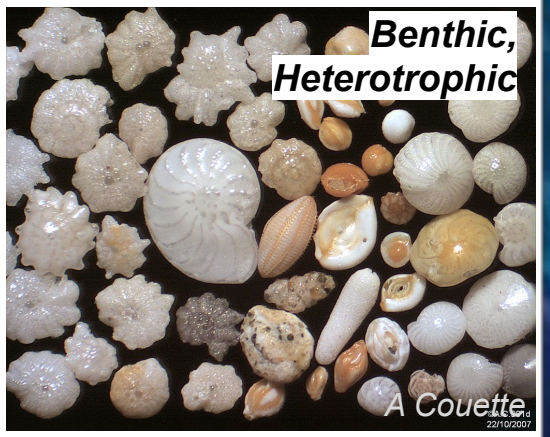
Fossil of the Week: FORAMINIFERA

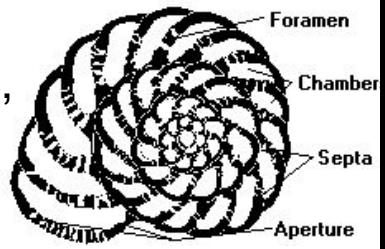
Who: Single celled amoeboid eukaryotes

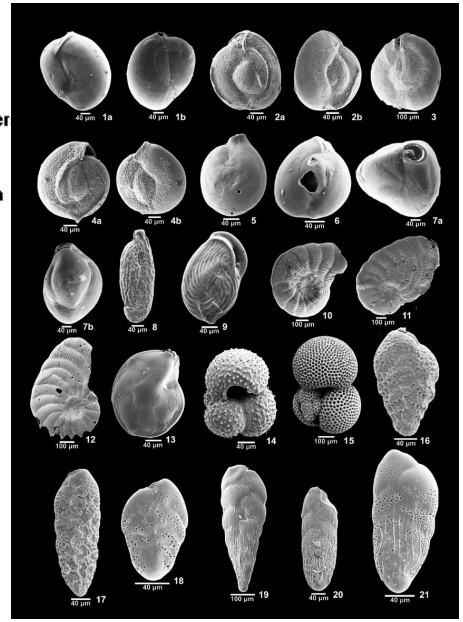
What: Calcium carbonate “test” readily forms microfossils, chambers recording growth stages

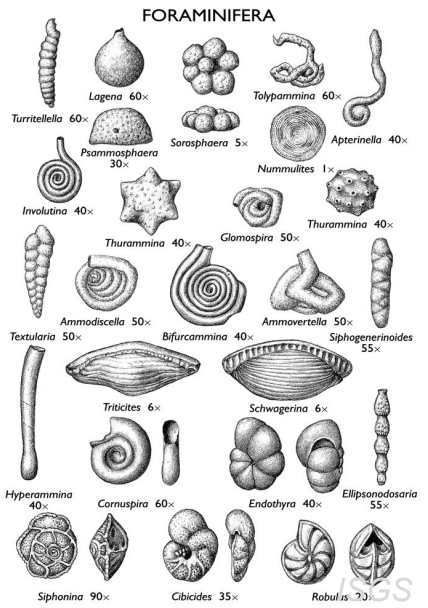
When: Phanerozoic, more diverse from Jurassic onwards

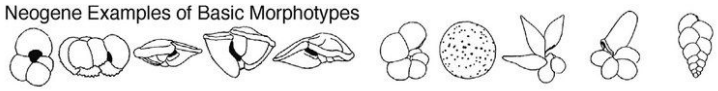


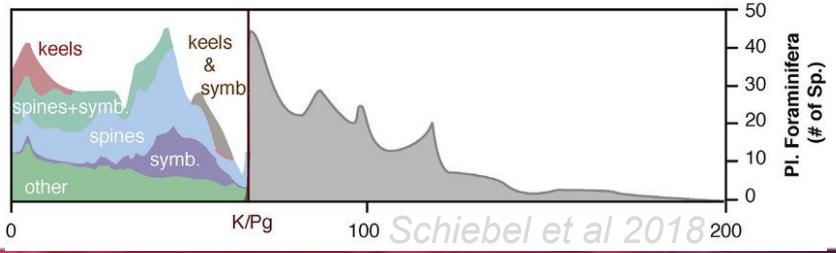




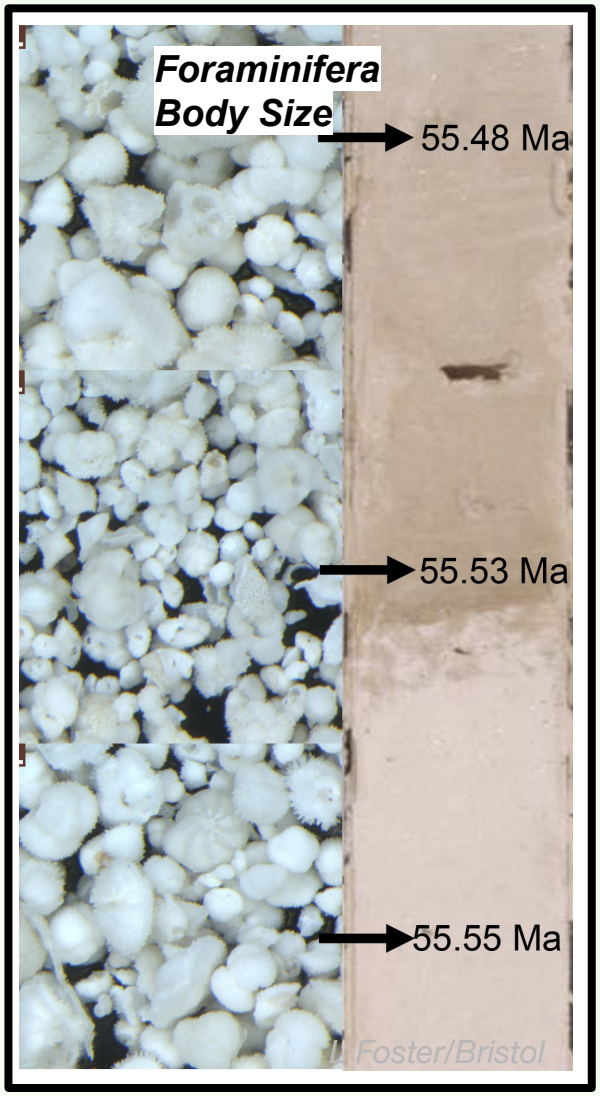


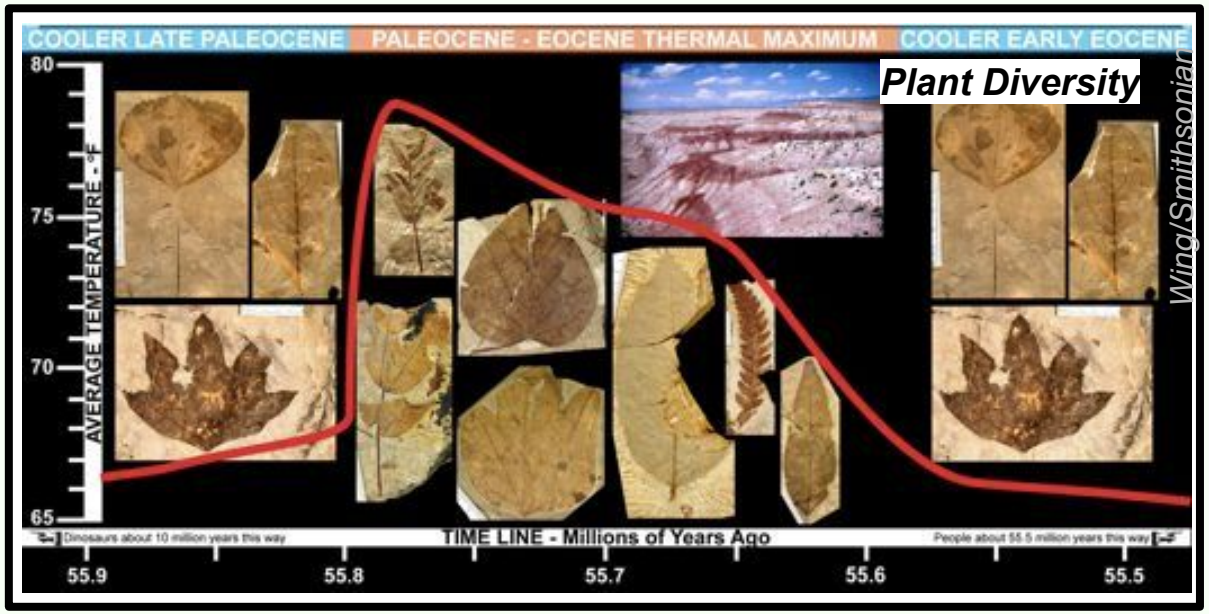


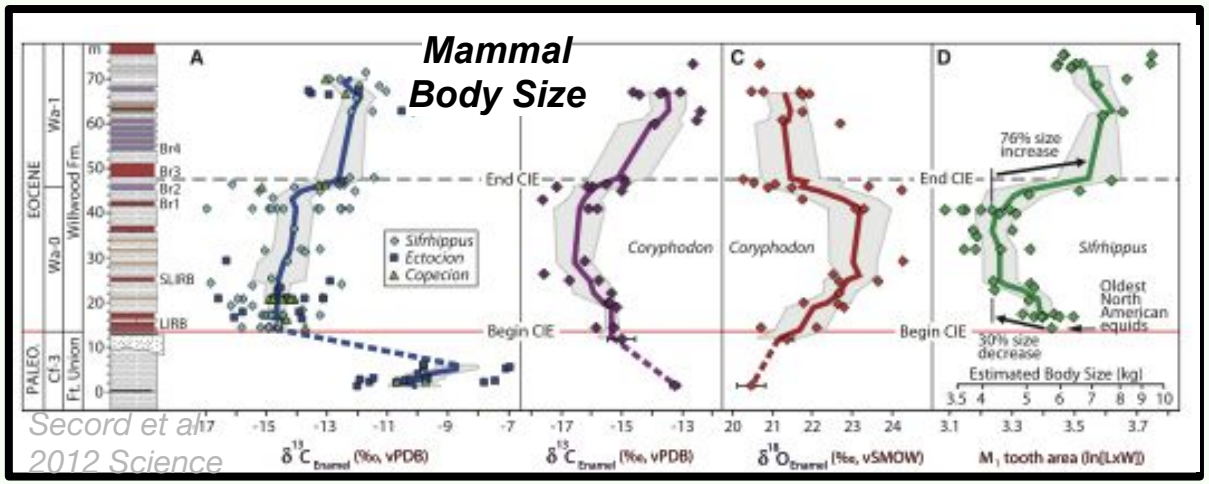




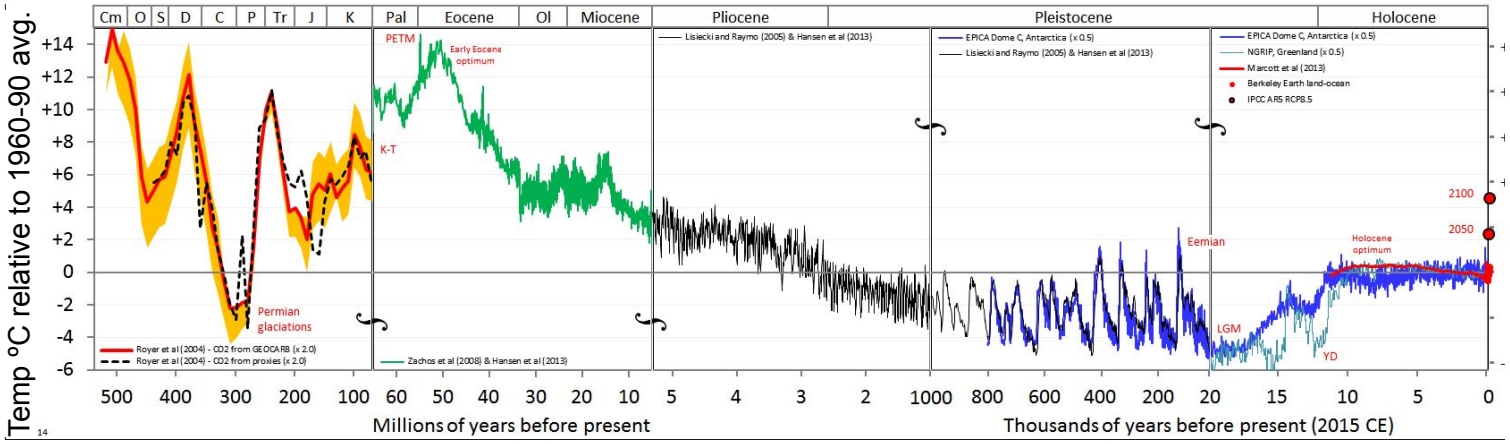
PETM Biotic Responses



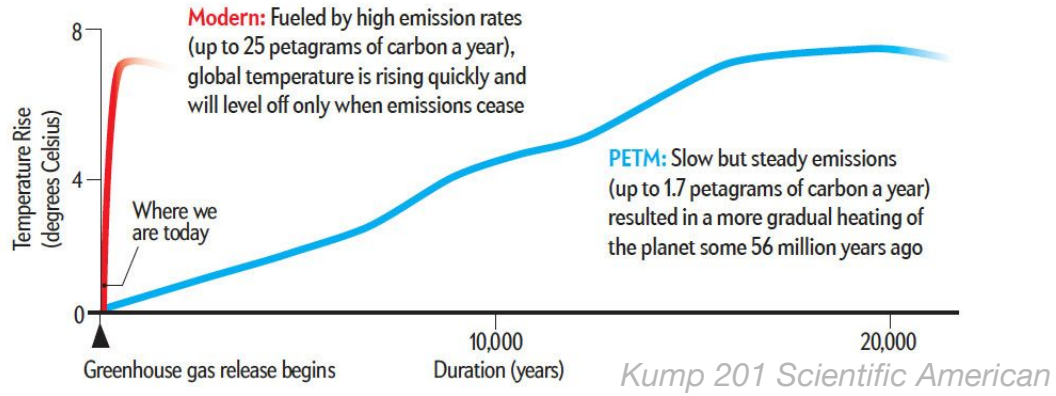




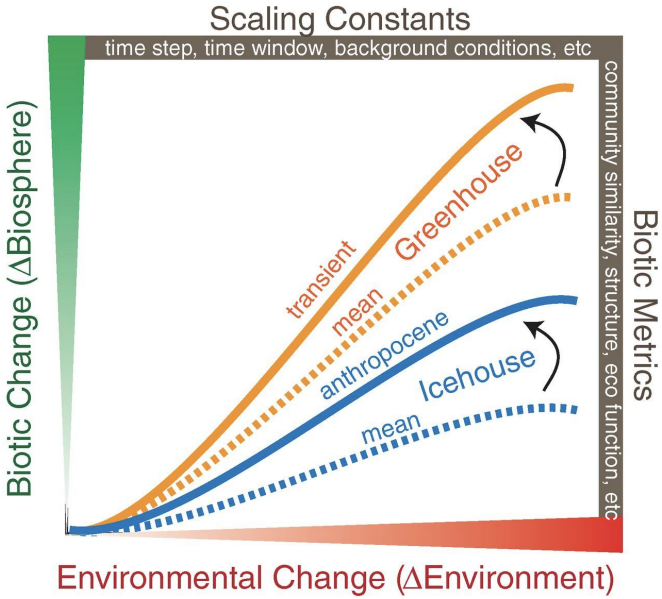
PETM a Proxy for Modern Climate Change?



PETM highest temp for Phan-erozoic



Kump 201 Scientific American



Norris et al 2013 Science

Rate of emissions and temp rise much higher today than even at the PETM

Modern earth system might be more resilient, some ecosystems are at more risk

Palaeogene Summary

- Rapid diversification of vertebrates following the end Cretaceous extinction event
- Very warm conditions toward the Eocene



Palaeogene, Dawn of the Modern

Climate and Evolution

Intended Learning Objective: Compare modern climate change with the Palaeocene-Eocene Thermal Maximum in terms of drivers, responses, scale, and rates of change

Now take a look at associate task/discussion

Robert Sansom, U. Manchester



Part 9
Palaeogene, Dawn of the Modern

Neogene and Hoofed Mammals

EART11100: How to Grow a Planet
Robert Sansom
University of Manchester

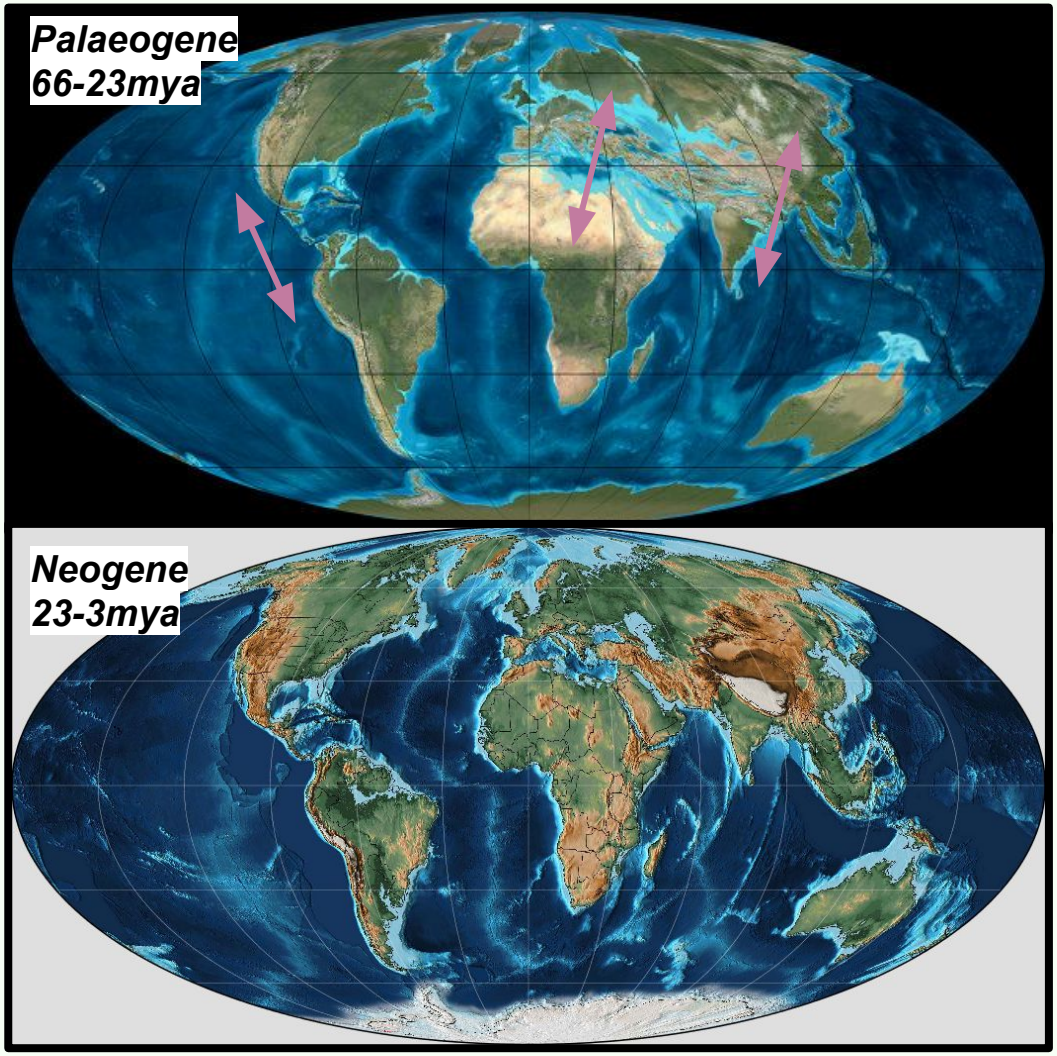


Palaeogene Summary

- Diversification of vertebrates after the end Cretaceous extinction
- Warm conditions in the Eocene, particularly the thermal maximum
- Spread of grasses



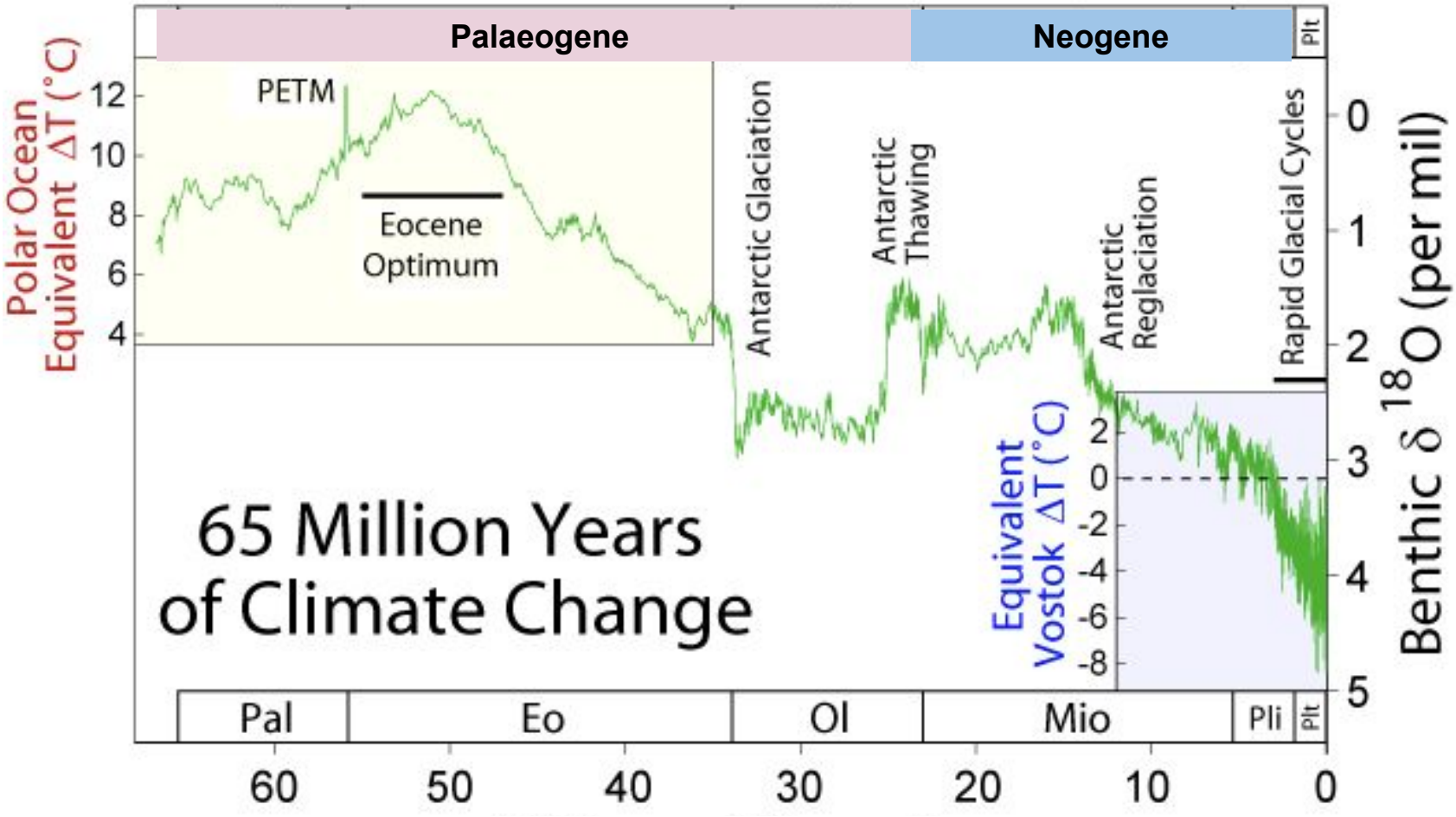
The Neogene



Continents and planet finally resemble modern arrangement

- Isthmus of Panama forms
- Collision of Indian plate with Eurasia
- Contact between Africa and Eurasia
- Ice-covered Poles return

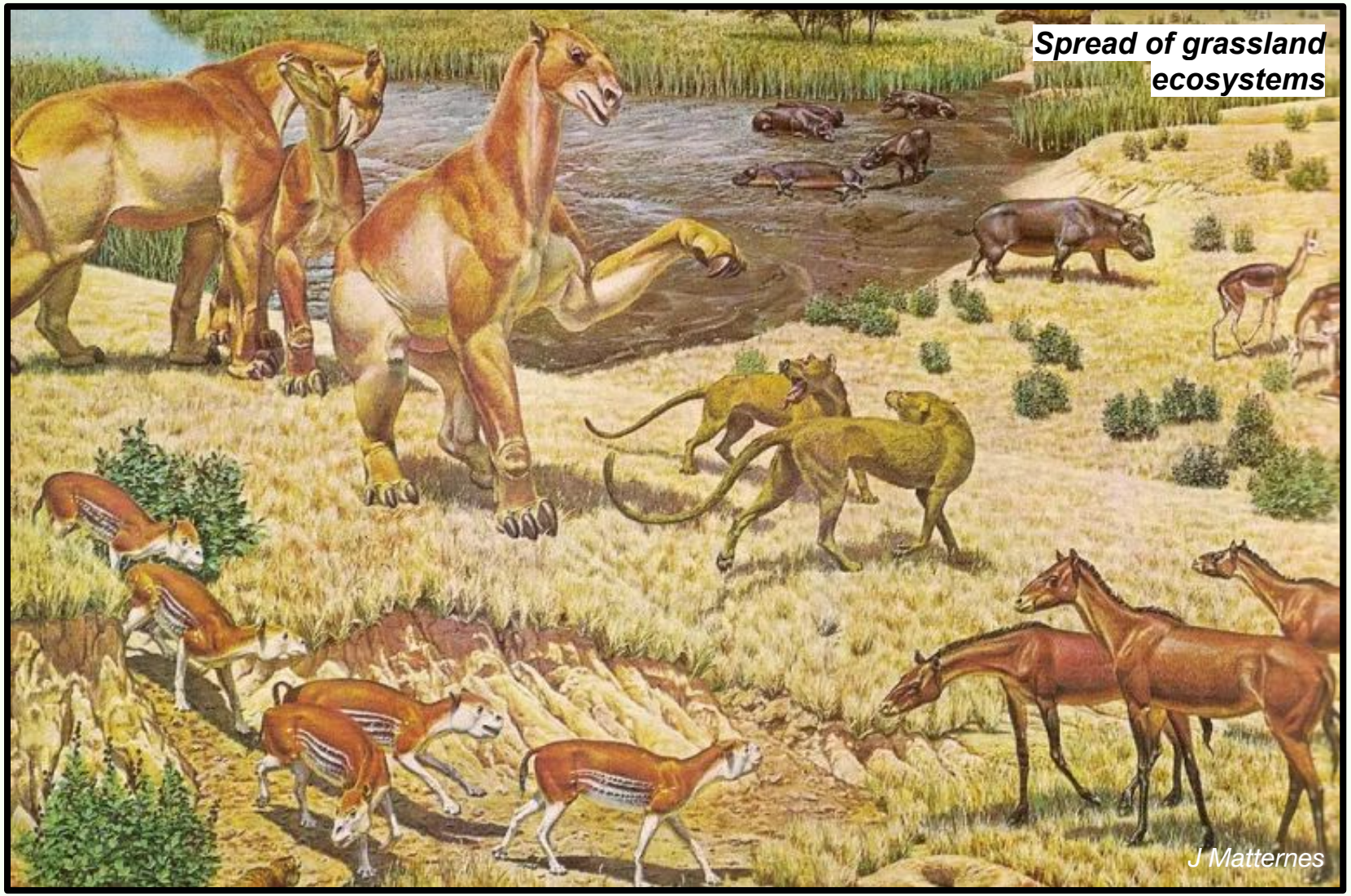
Neogene Climate



R Rohde WikiCommons

Cooler global temperatures, closer to modern range. Seasonality and glaciations

Neogene Climate



Browsing and Grazing

Browsers are herbivores that feed on leaves, shoots and fruits of high growing woody plants (e.g. shrubs)



Grazers are herbivores that feed on the low vegetation, particularly grasses (but also algae)



Ungulates: Hoofed Mammals



Even toed ungulates



Odd toed ungulates

Wikimedia Commons

Ungulates are diverse mostly **herbivorous** mammals

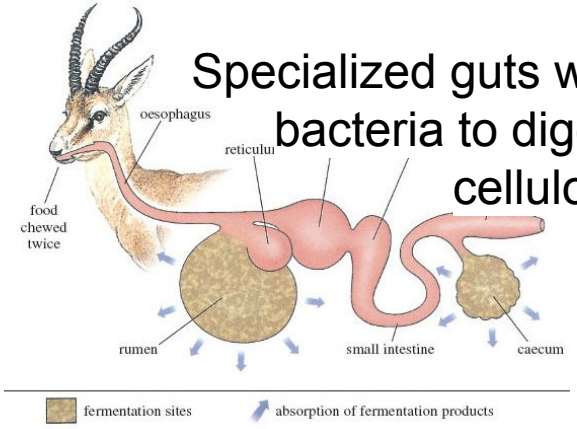


A **hoof** is the tip of toe strengthened with keratin

Encyc. Britanica

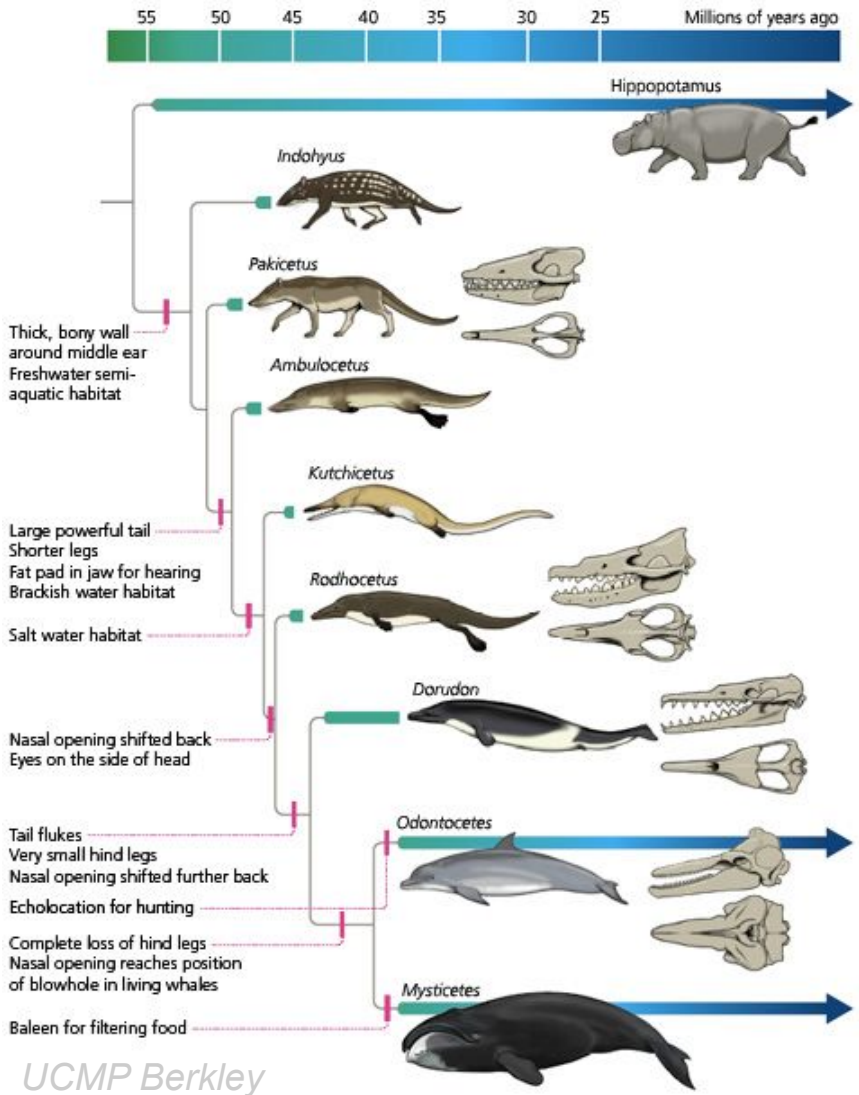


Specialized **molars** for grinding vegetation



Specialized guts with bacteria to digest cellulose

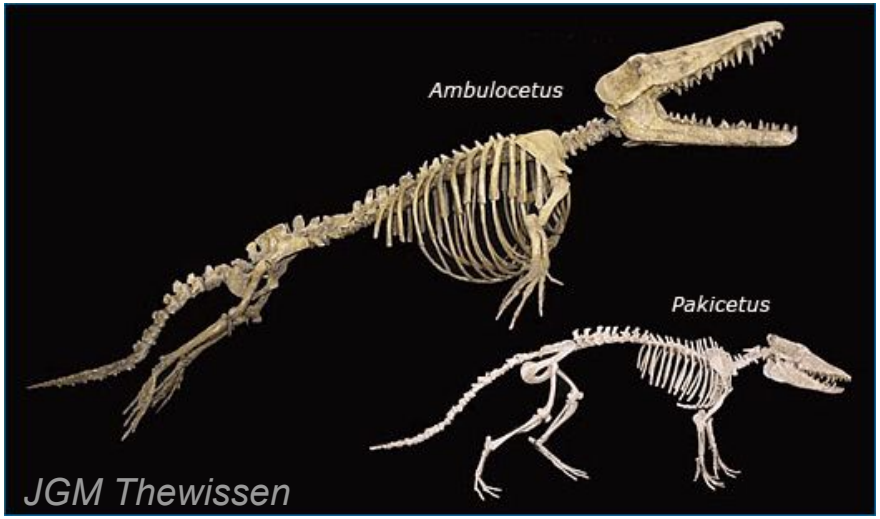
Whales are ‘hoofed’ mammals



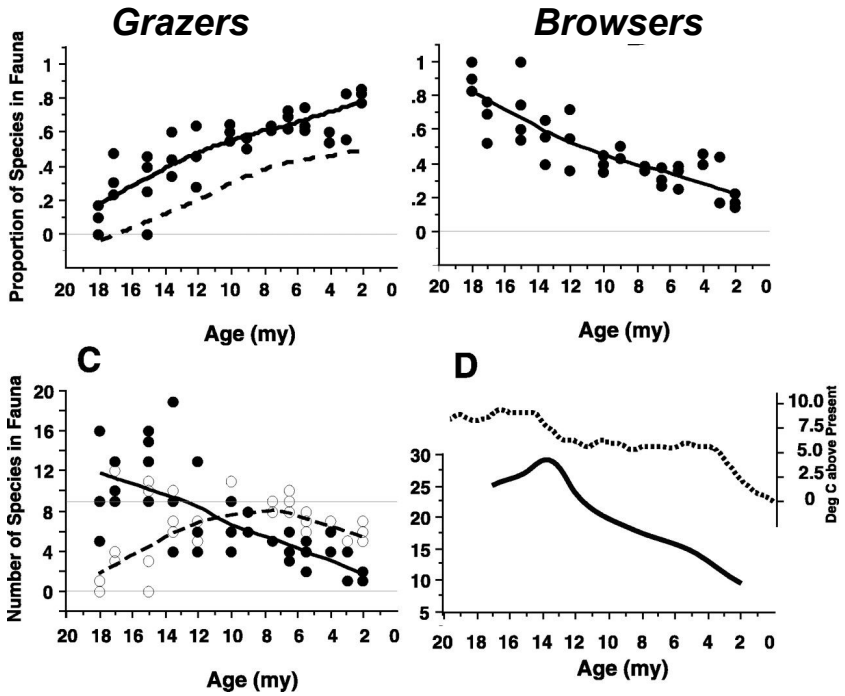
Molecule data confirmed that whales (**Cetacea**) are sister taxon to Hippos, firmly in the ungulate crown-group

A range of interested fossils reveal the transition from hoofed to fully aquatic

An example is *Ambulocetus*: a piscivorous walking whale



Neogene shift to Browsing



Janis et al 2000

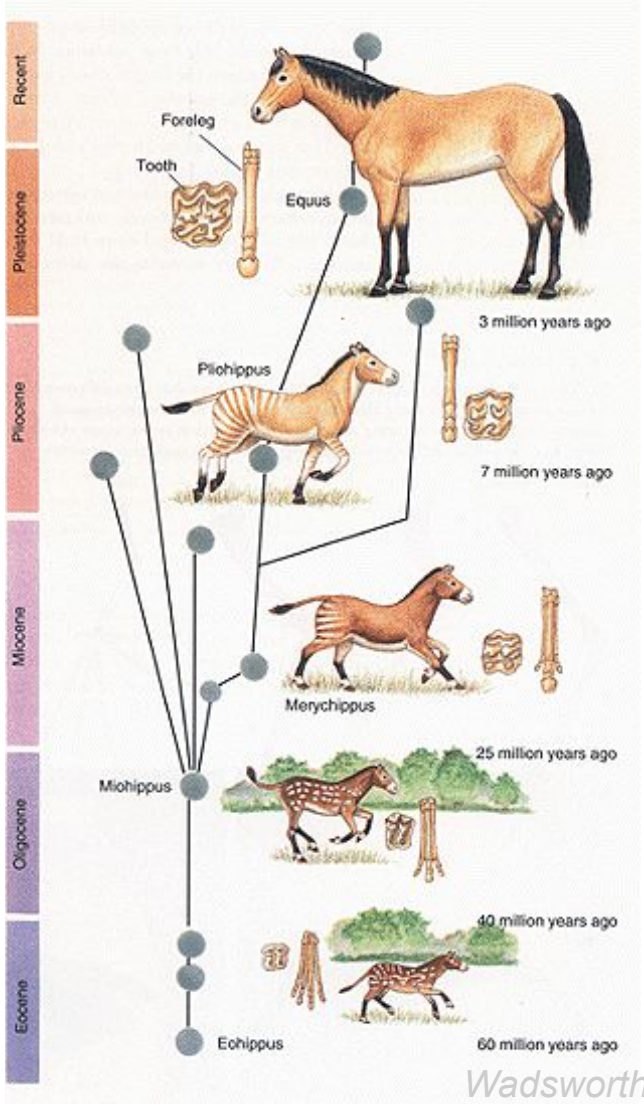
During the Neogene there was a shift from predominantly **browsing** ungulates to **grazing** ungulates

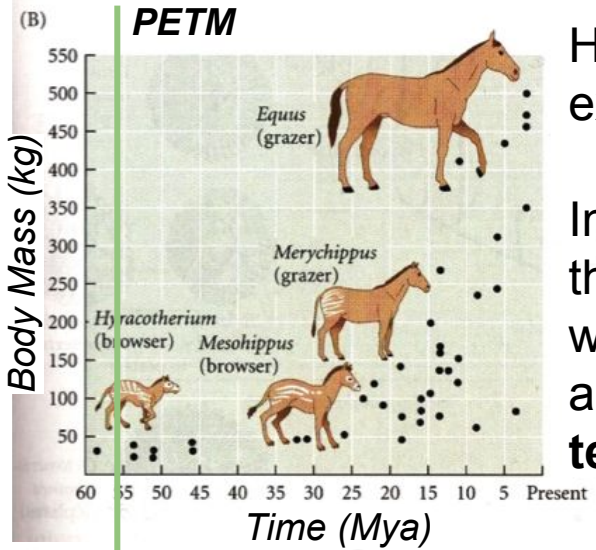




The spread of browsing would have had big ecological effect: the spread of grasslands (e.g. prairie, savannah)

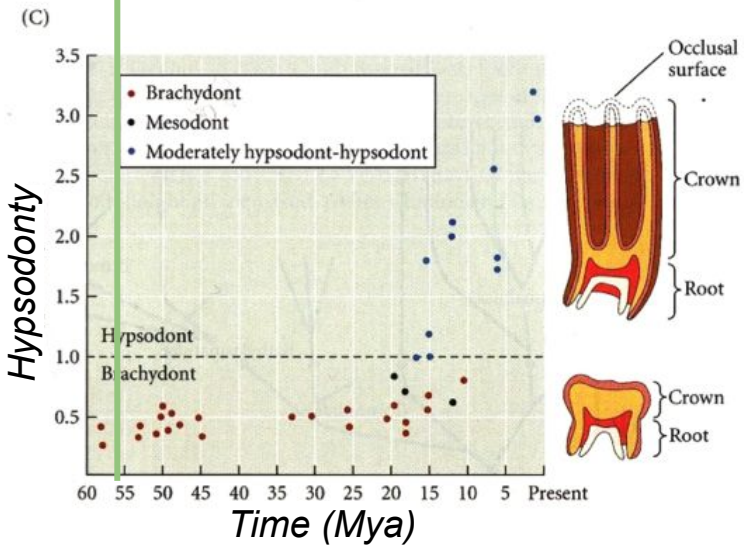
Horses and Body Size





Horses are a classic example of evolution

Increasing **body size** through time along with reduction of toes and specialization of **teeth** (hypsodonty)

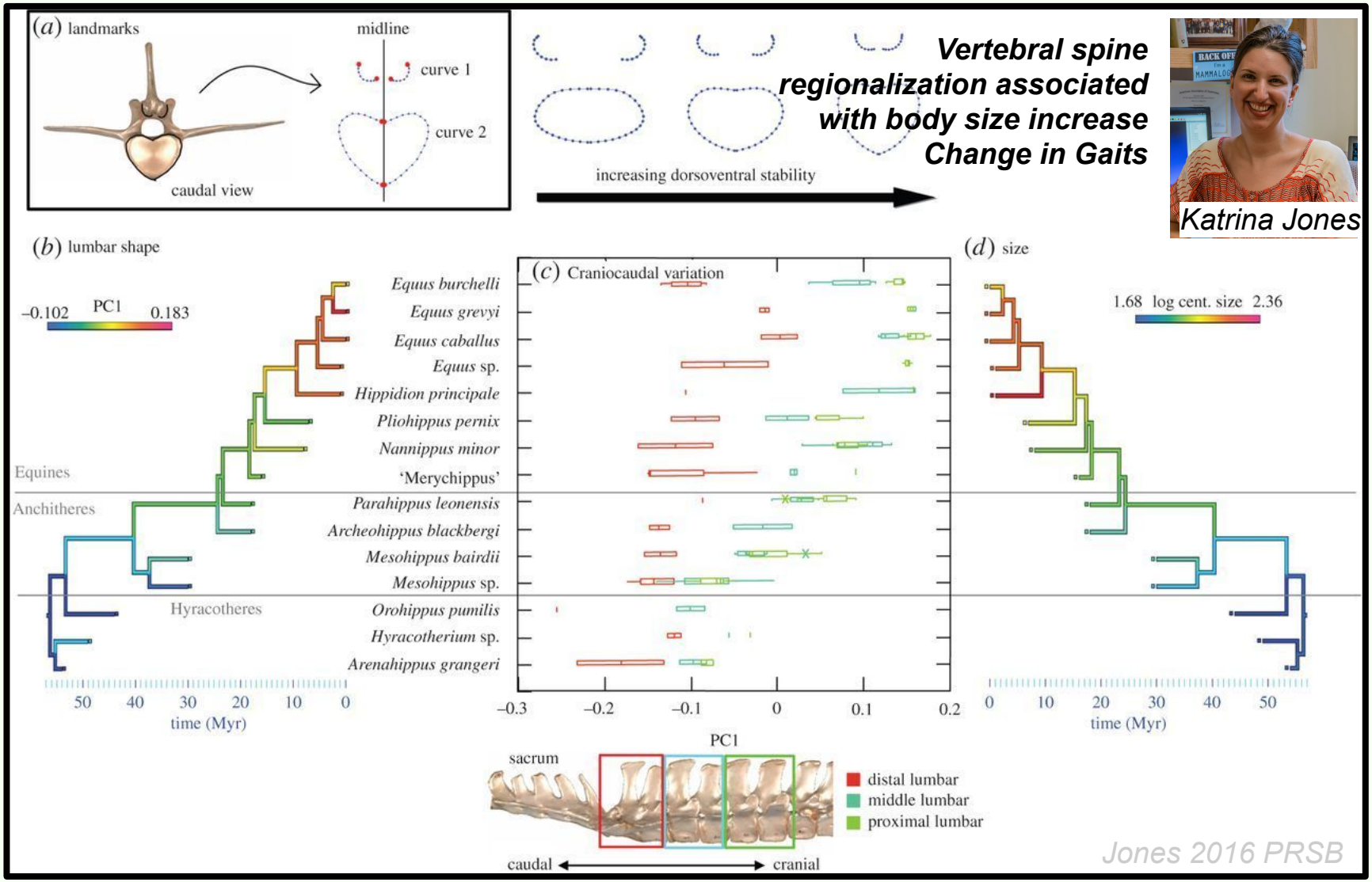


Hypsodont molar tooth

Brachydont molar tooth

Sinaeur

Horses and Body Size

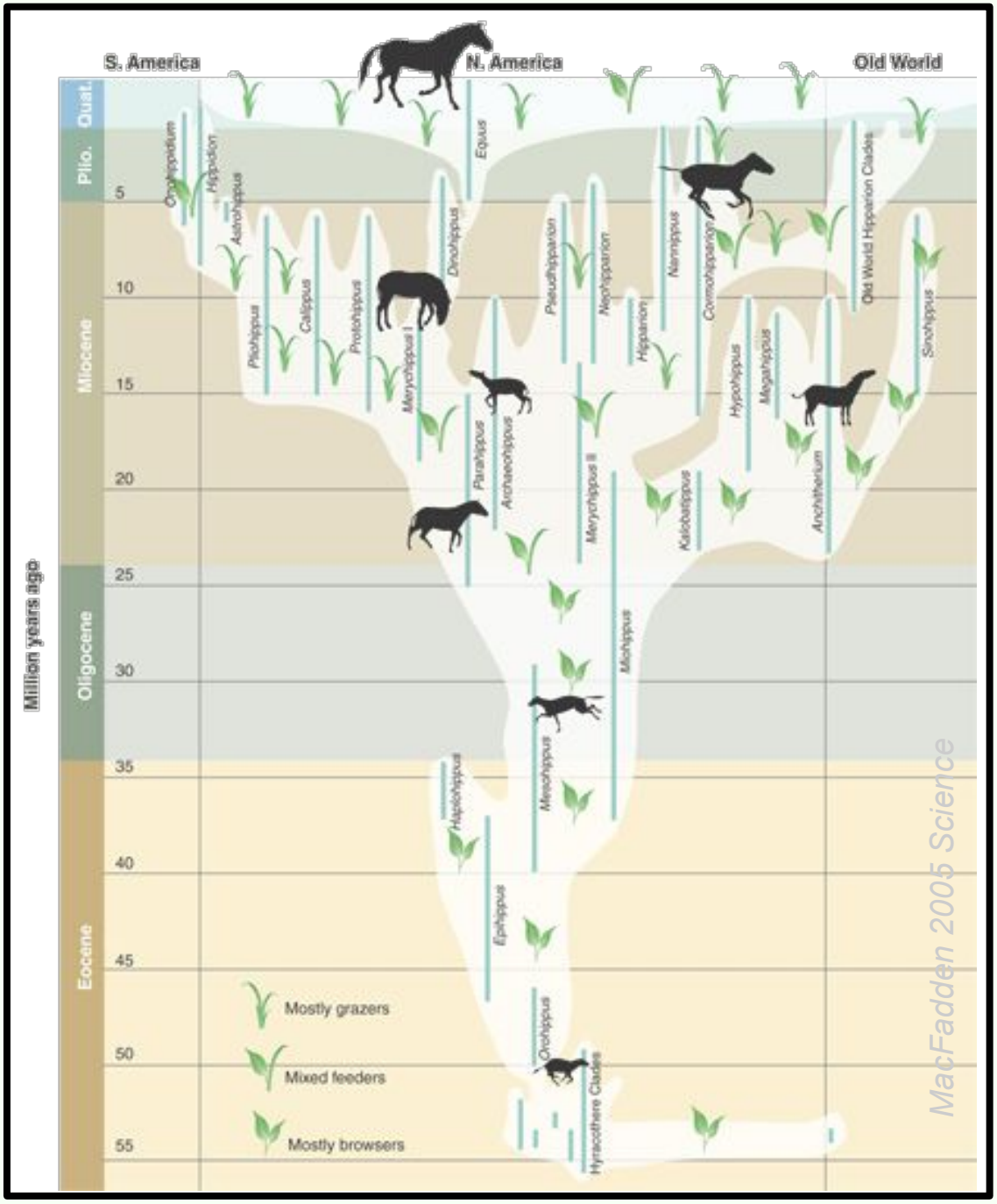


Katrina Jones

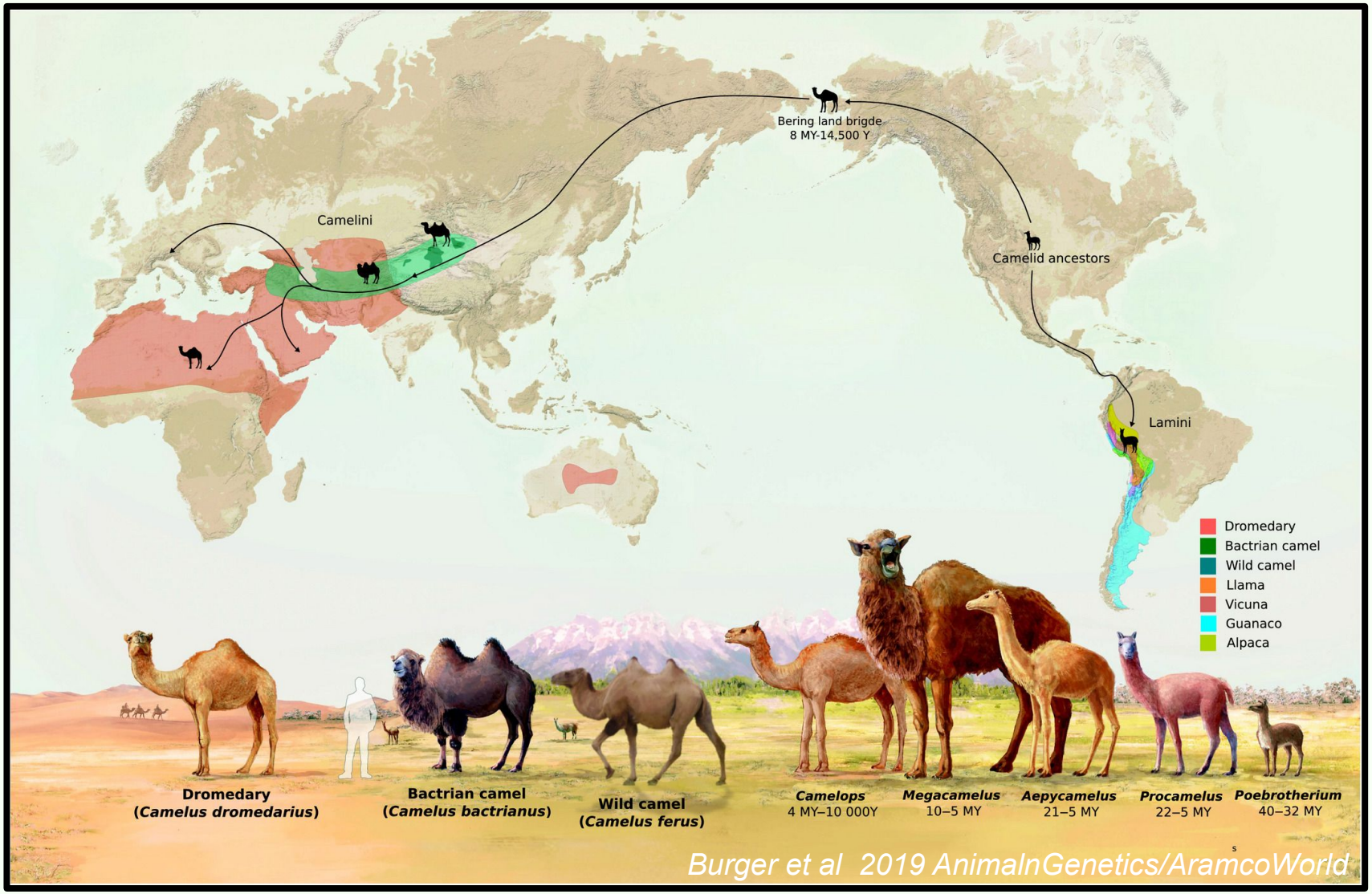
Jones 2016 PRSB

Horse Evolution

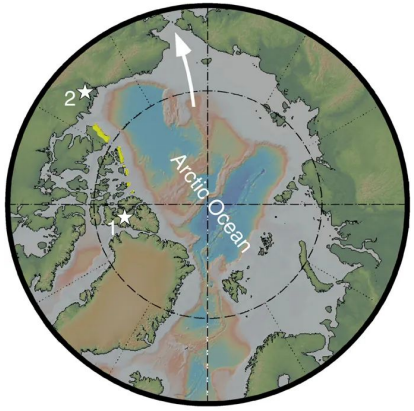
- Increasing body size
- Geographic Spread
- Shift from browsing to grazing

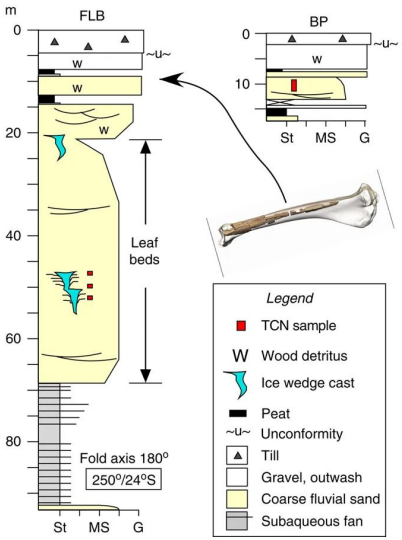


Camel Palaeobiogeography



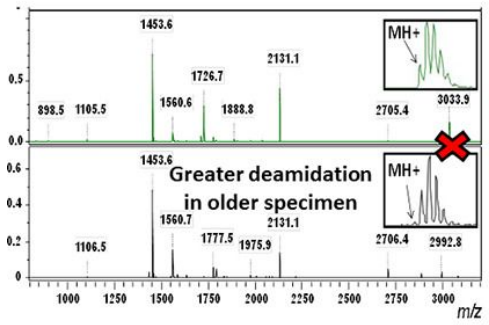
Fossil proteins





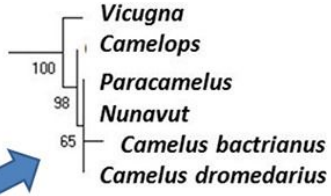
Bone fragment from Arctic Canada, unknown affinity

Rybczynski et al Nat Com 2013
Buckley et al 2019 J Proteomics



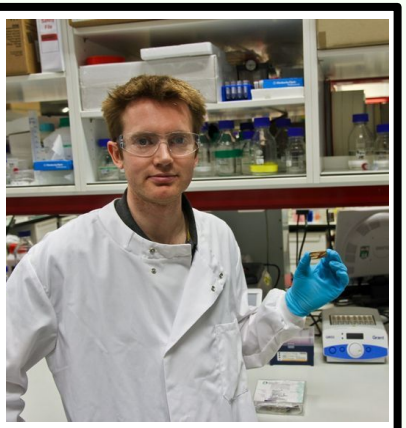
Diagenetic modifications result in few missing peptides





Omission of a single peptide results in phylogenetic topology change

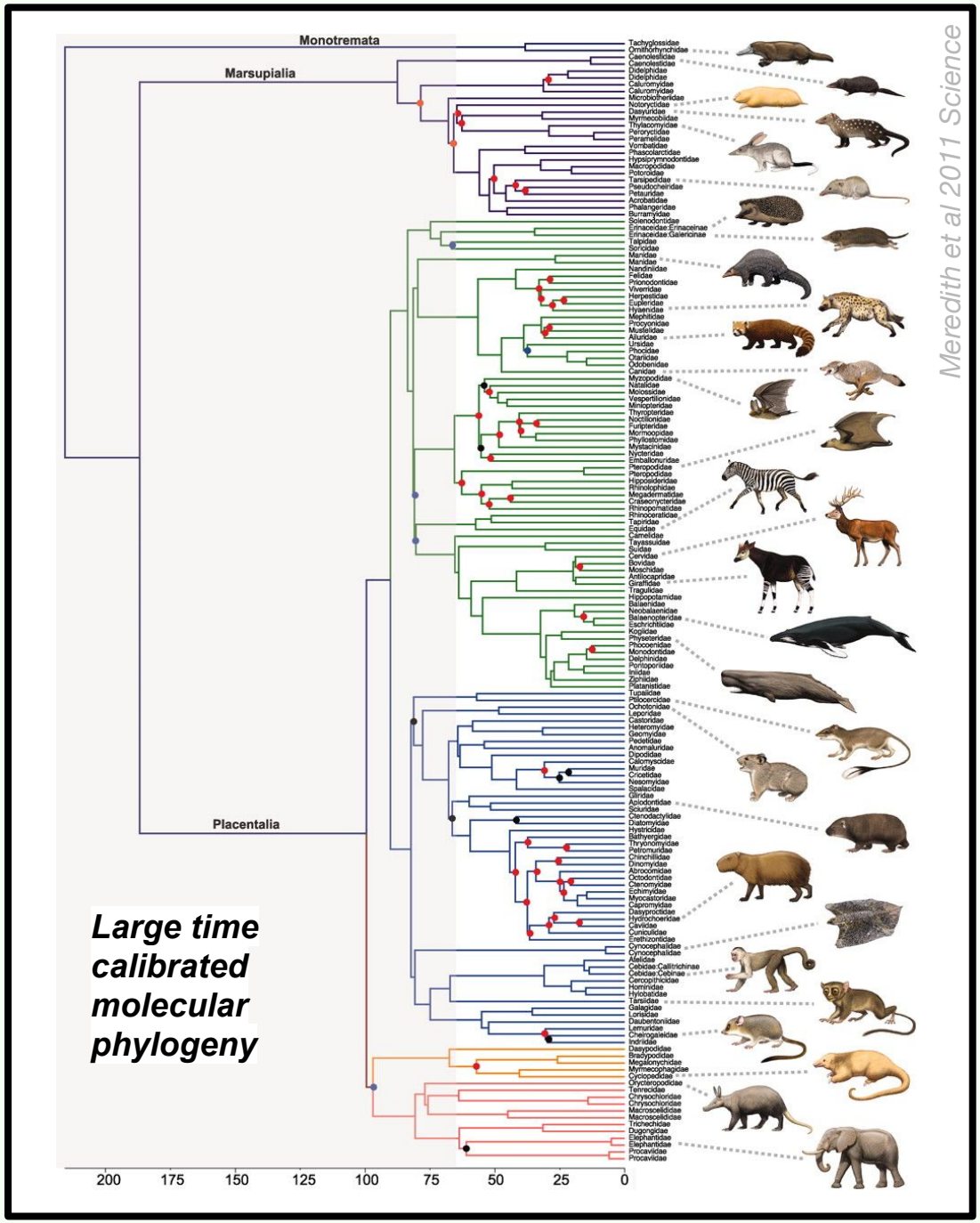






Camels migrated to North America via Arctic during warm period, 3.5 million years ago

Mammal Evolution



Palaeogene, Dawn of the Modern

Neogene and Hoofed Mammals

Intended Learning Objective: Describe the evolution and ecology of hoofed mammals with reference to earth history

Now take a look at associate task/discussion

Robert Sansom, U. Manchester

